

BFS Traversal

Step 1: Start at A

Legend

Complete

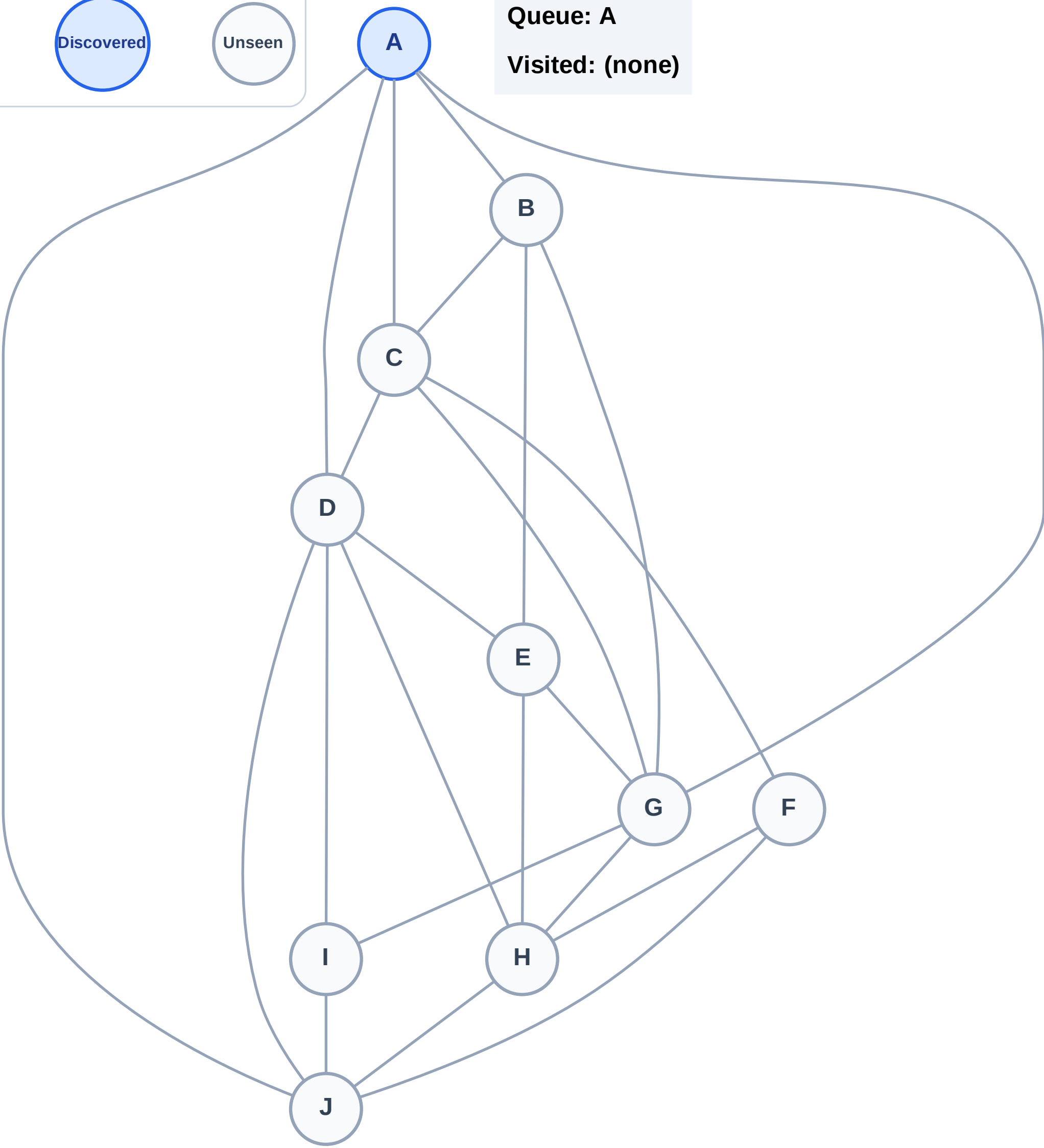
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

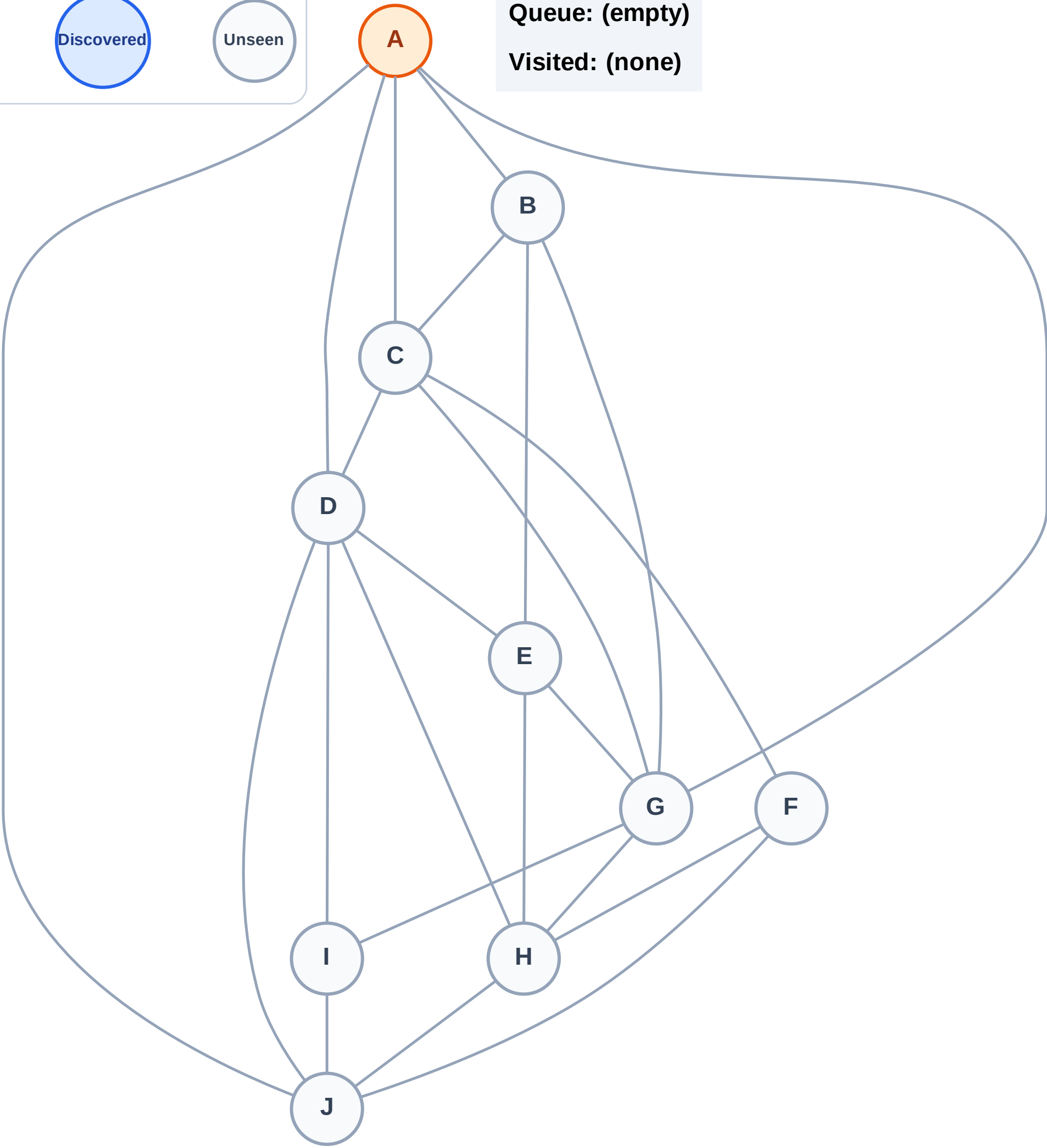
Complete

Processing

Discovered

Unseen

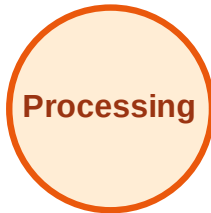
Queue: (empty)
Visited: (none)



BFS Traversal

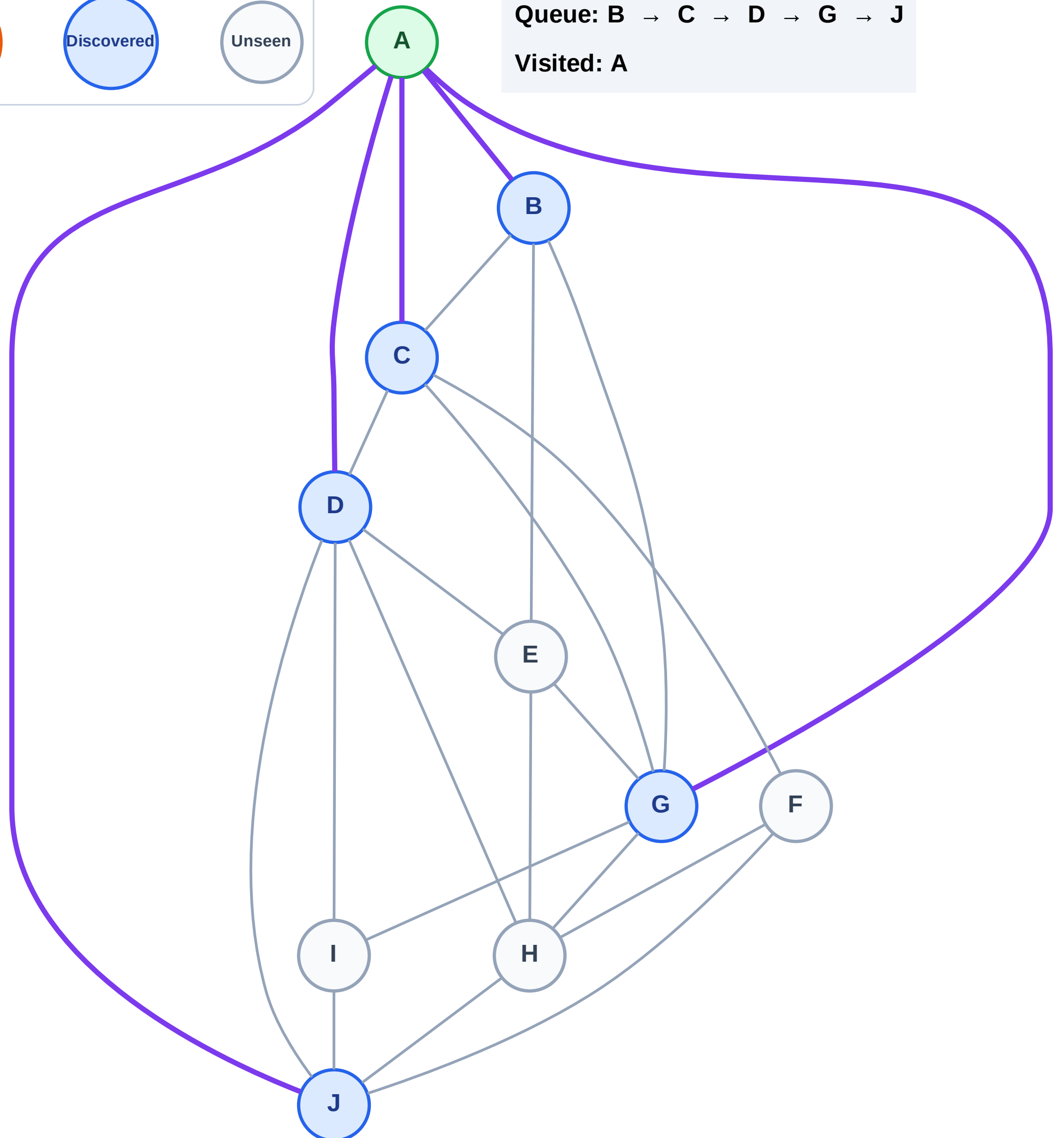
Step 3: Discover B, C, D, G, J from A

Legend



Queue: B → C → D → G → J

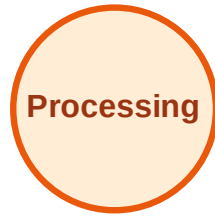
Visited: A



BFS Traversal

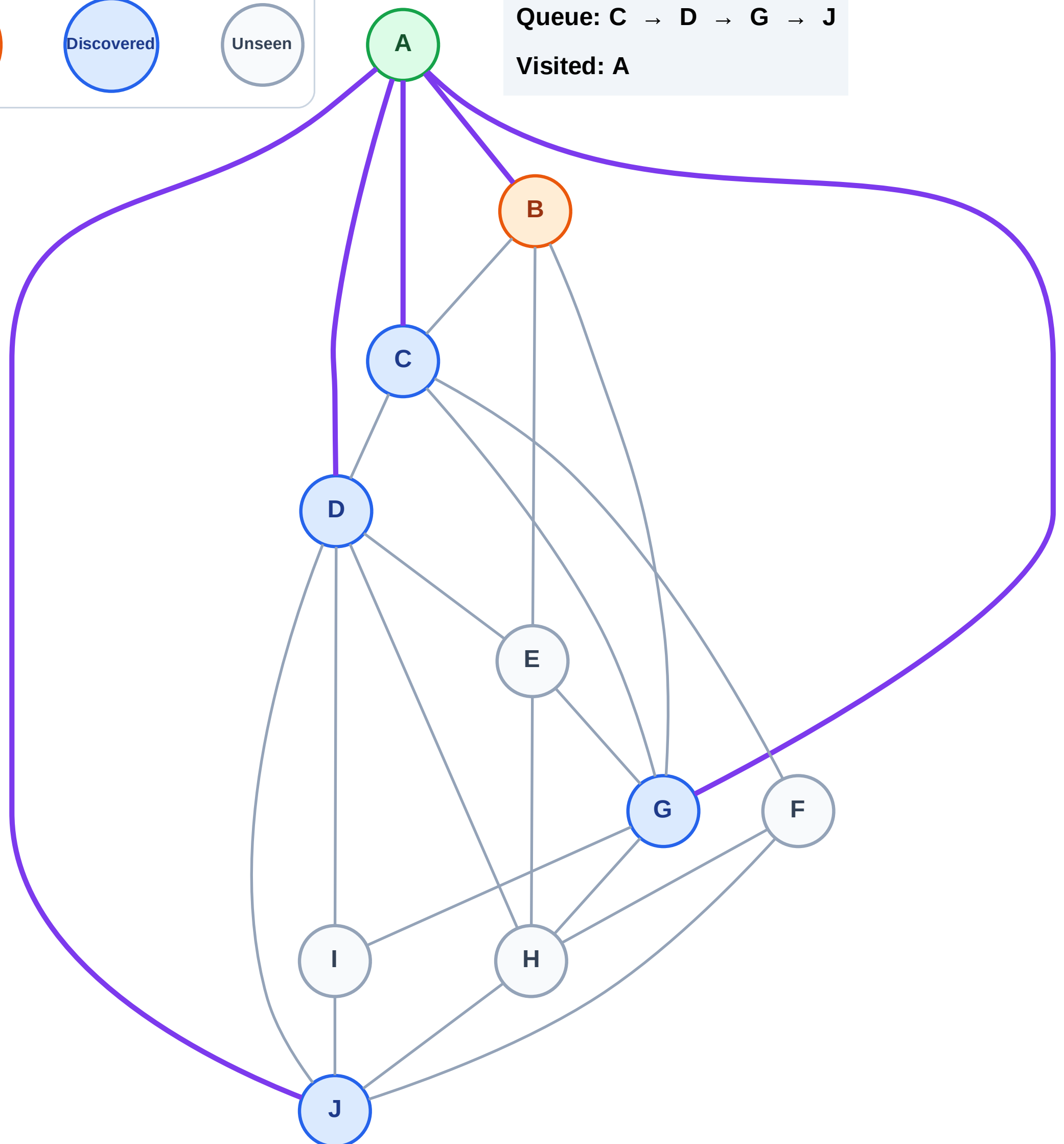
Step 4: Process node B

Legend



Queue: C → D → G → J

Visited: A



BFS Traversal

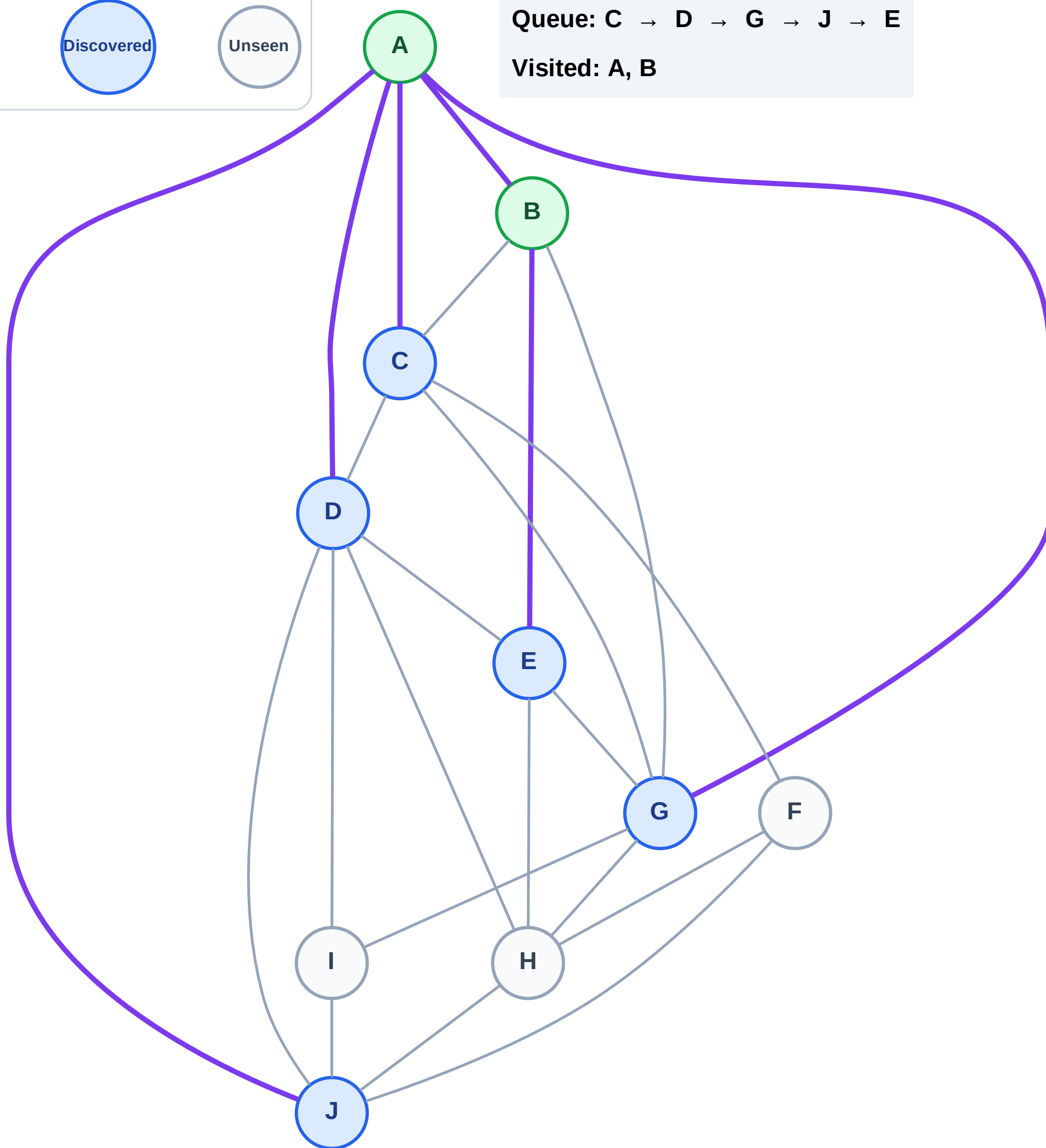
Step 5: Discover E from B

Legend



Queue: C → D → G → J → E

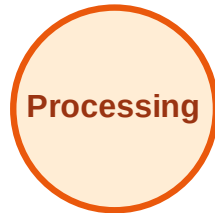
Visited: A, B



BFS Traversal

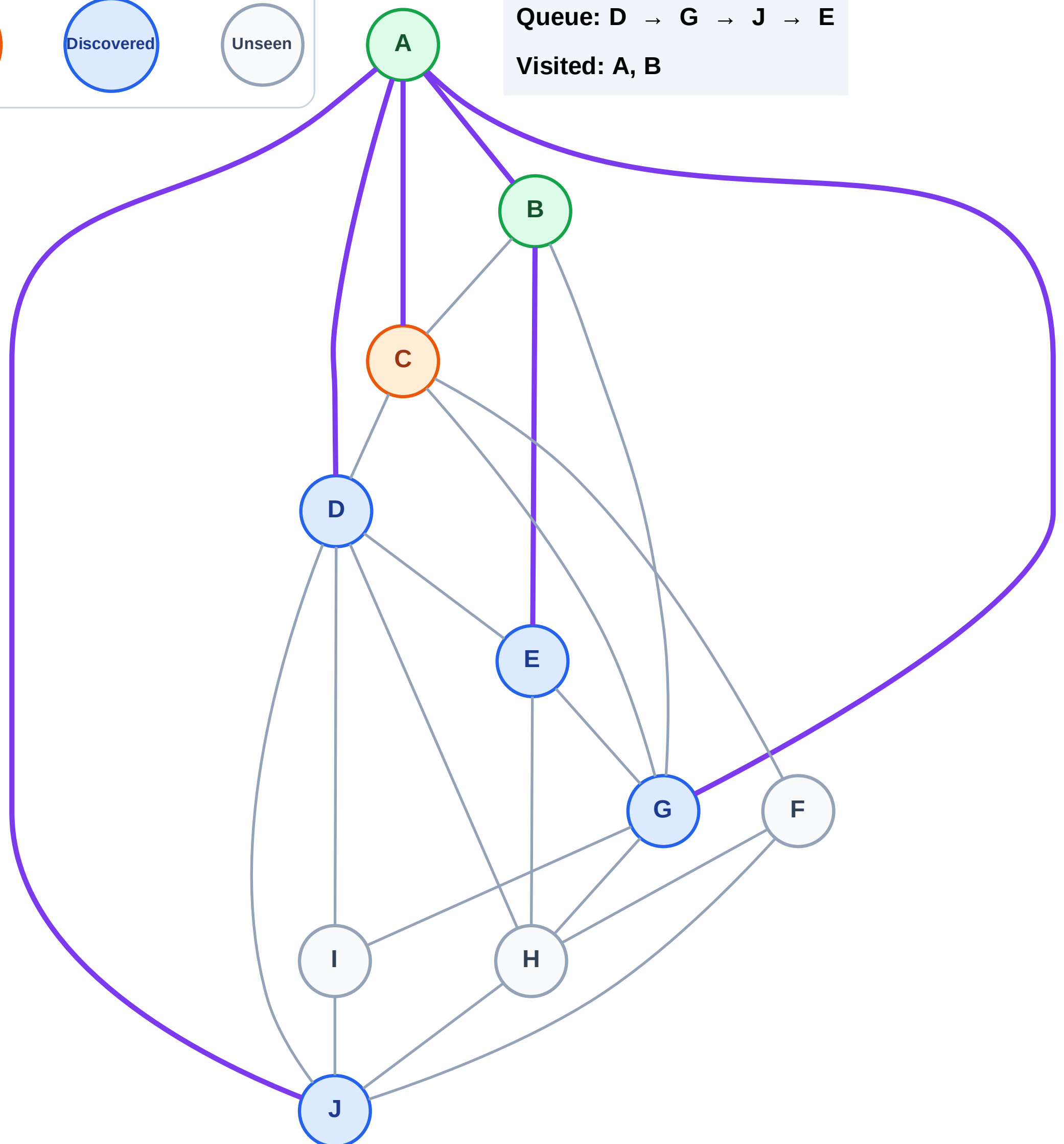
Step 6: Process node C

Legend



Queue: D → G → J → E

Visited: A, B



BFS Traversal

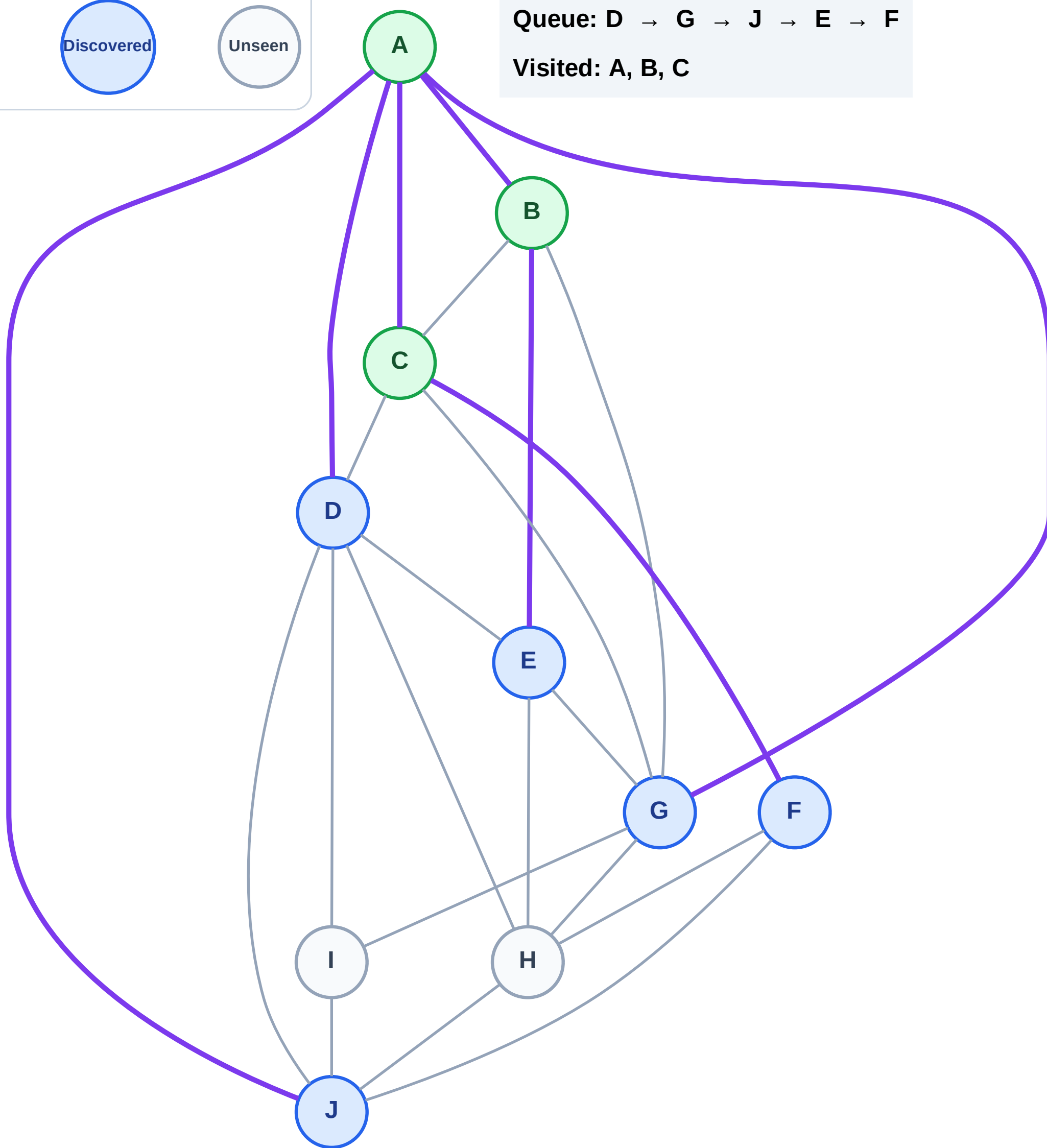
Step 7: Discover F from C

Legend



Queue: D → G → J → E → F

Visited: A, B, C



BFS Traversal

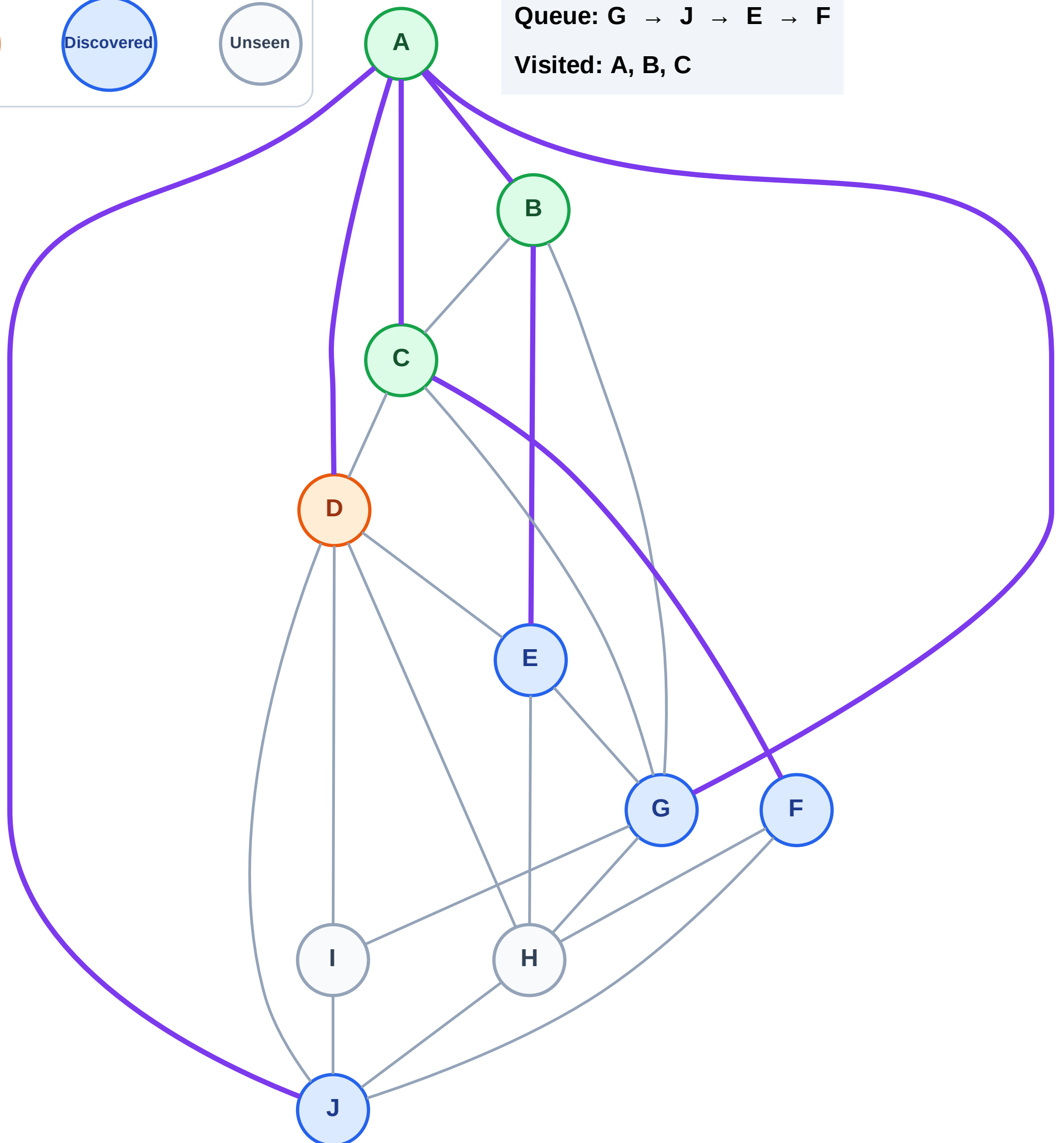
Step 8: Process node D

Legend



Queue: G → J → E → F

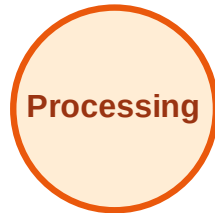
Visited: A, B, C



BFS Traversal

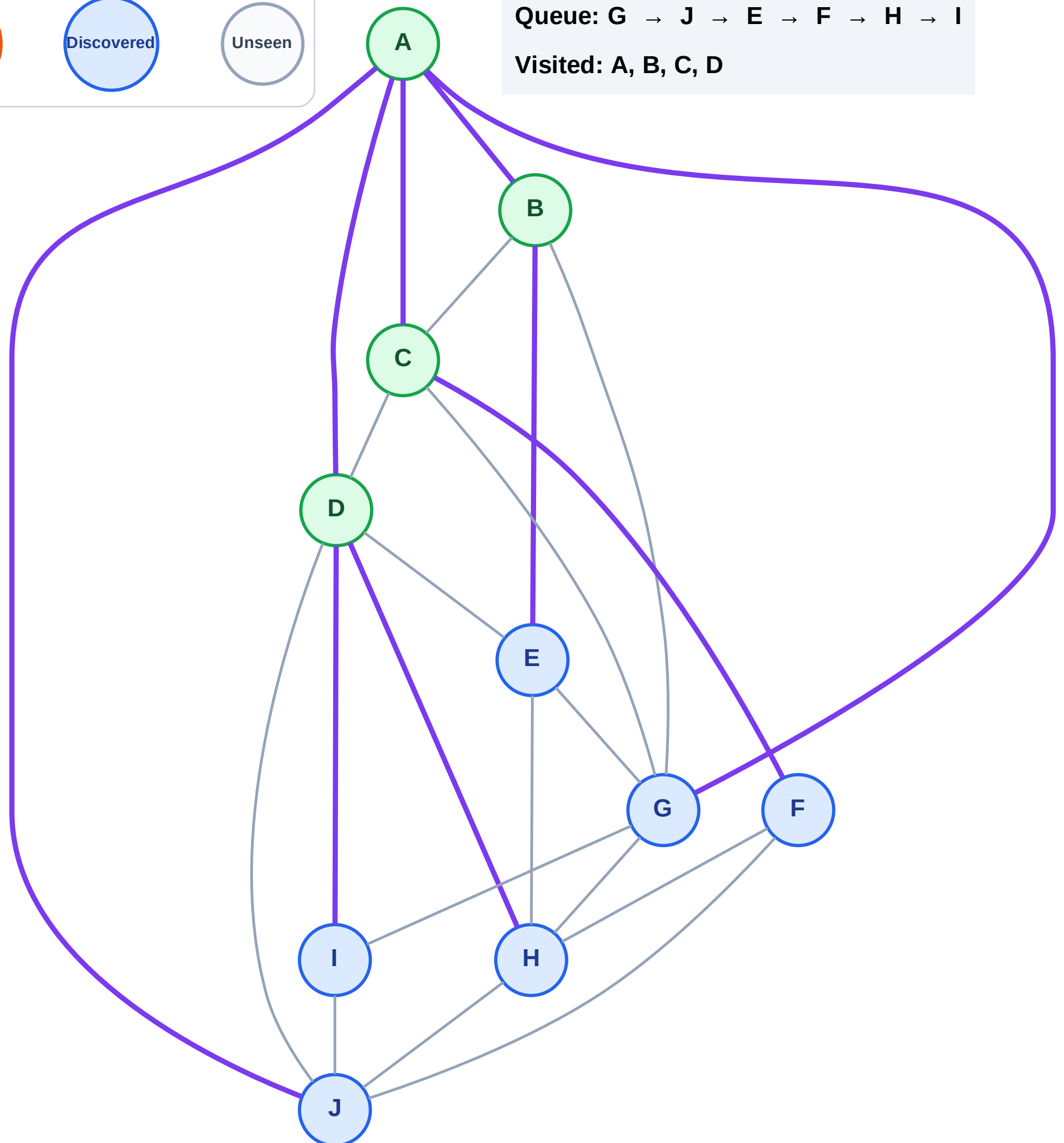
Step 9: Discover H, I from D

Legend



Queue: G → J → E → F → H → I

Visited: A, B, C, D



BFS Traversal

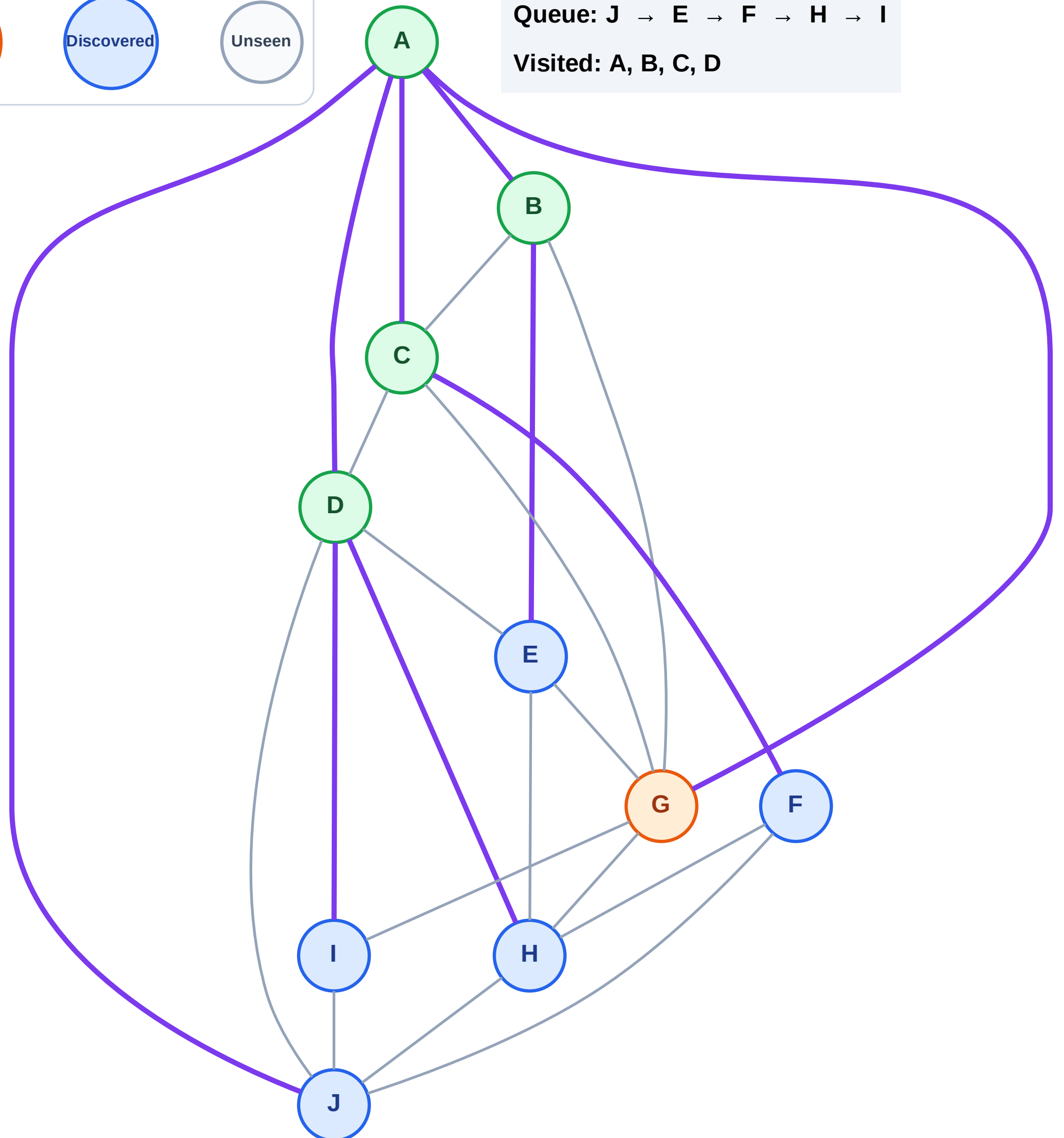
Step 10: Process node G

Legend



Queue: J → E → F → H → I

Visited: A, B, C, D



BFS Traversal

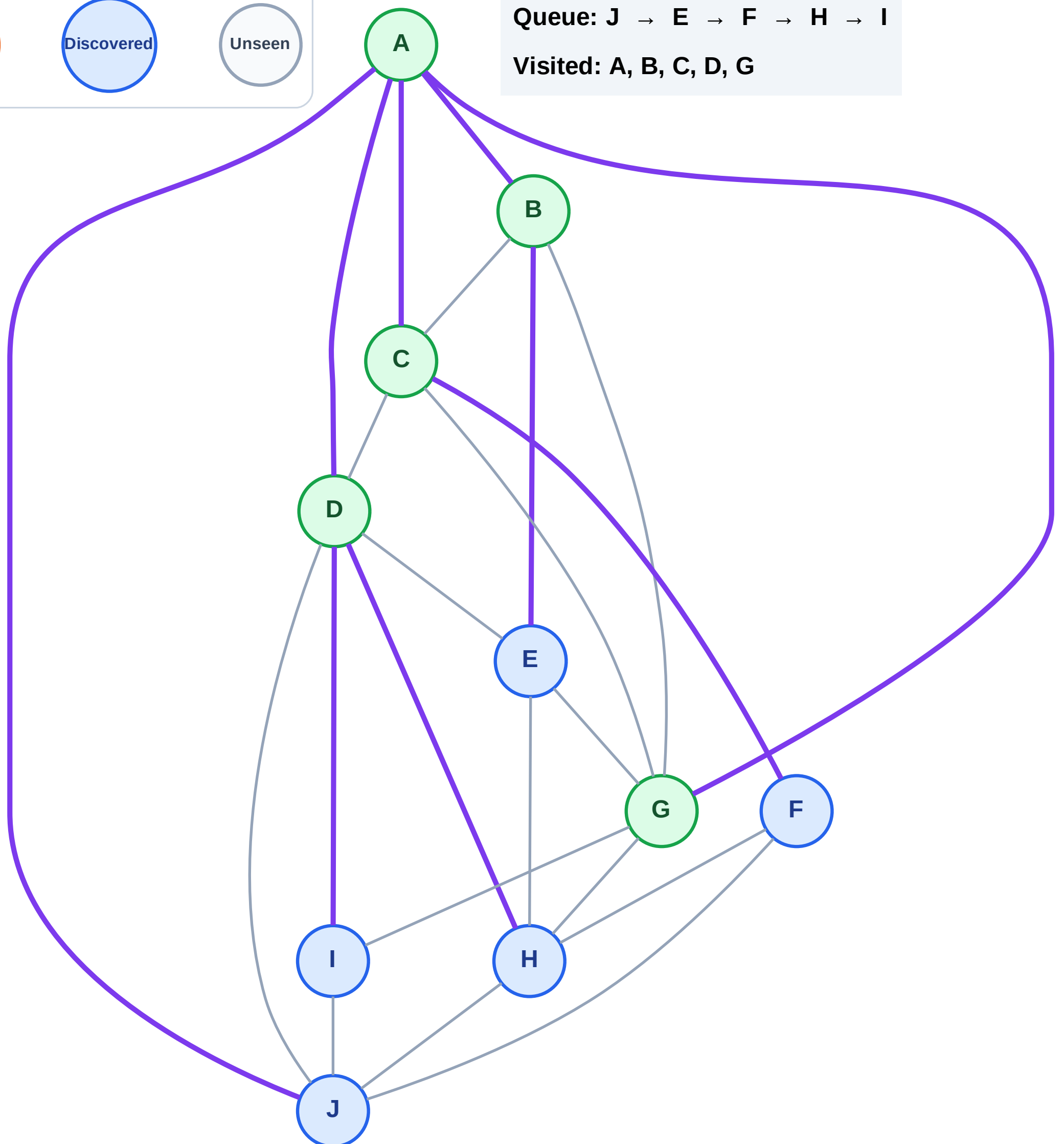
Step 11: No new neighbors from G

Legend



Queue: J → E → F → H → I

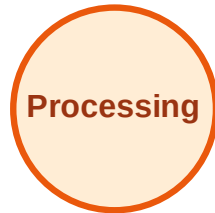
Visited: A, B, C, D, G



BFS Traversal

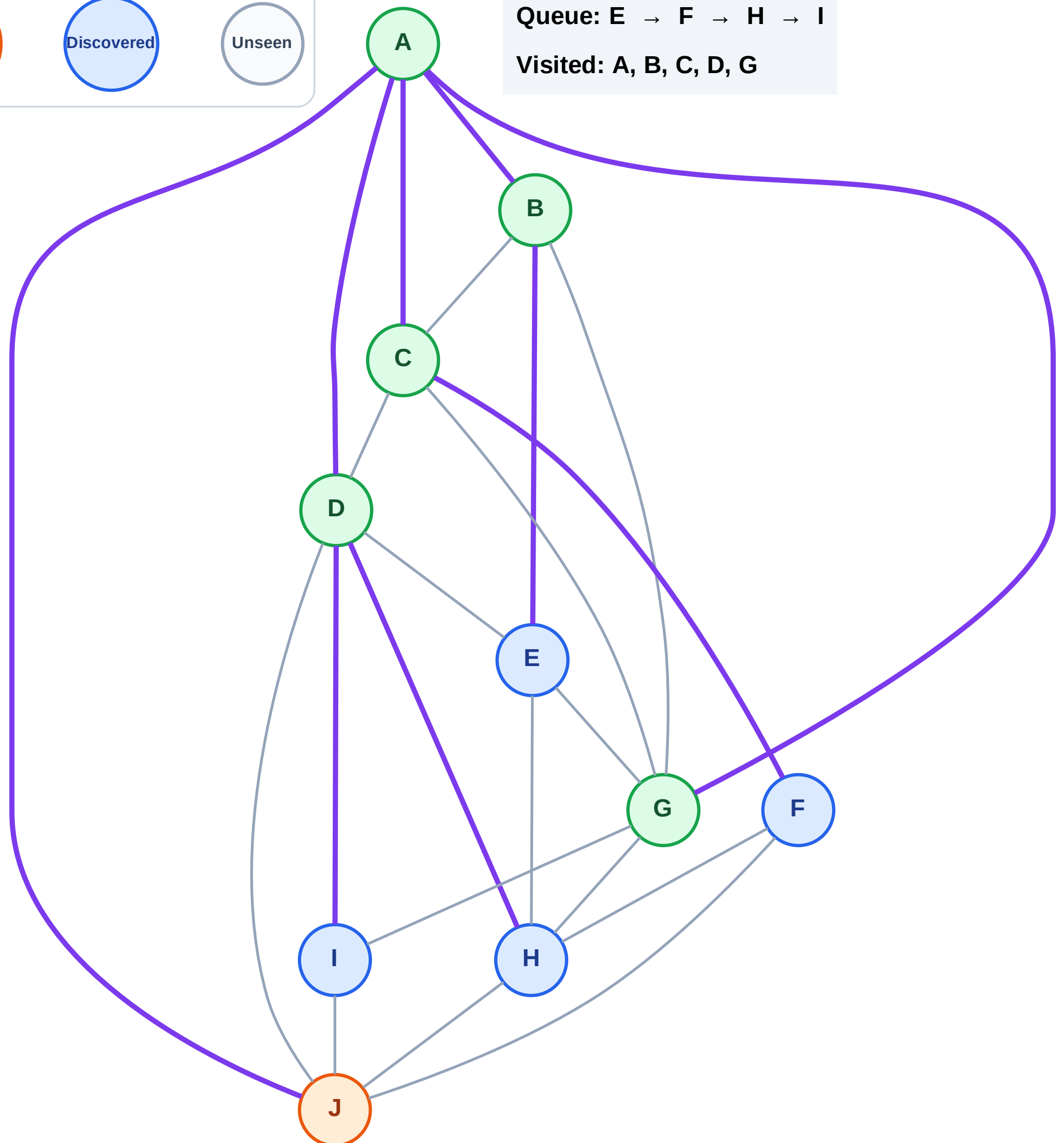
Step 12: Process node J

Legend



Queue: E → F → H → I

Visited: A, B, C, D, G



BFS Traversal

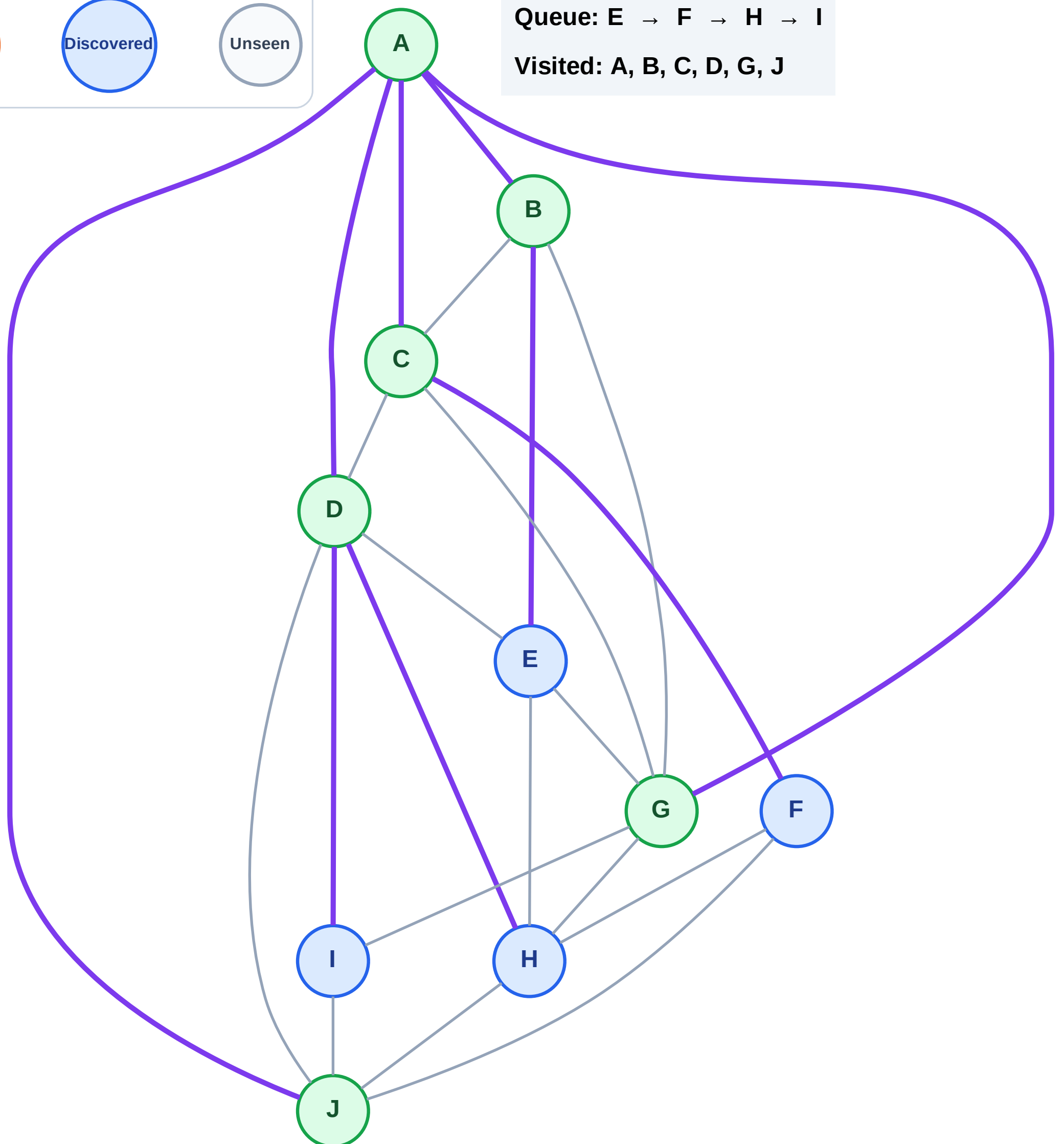
Step 13: No new neighbors from J

Legend



Queue: E → F → H → I

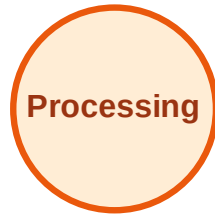
Visited: A, B, C, D, G, J



BFS Traversal

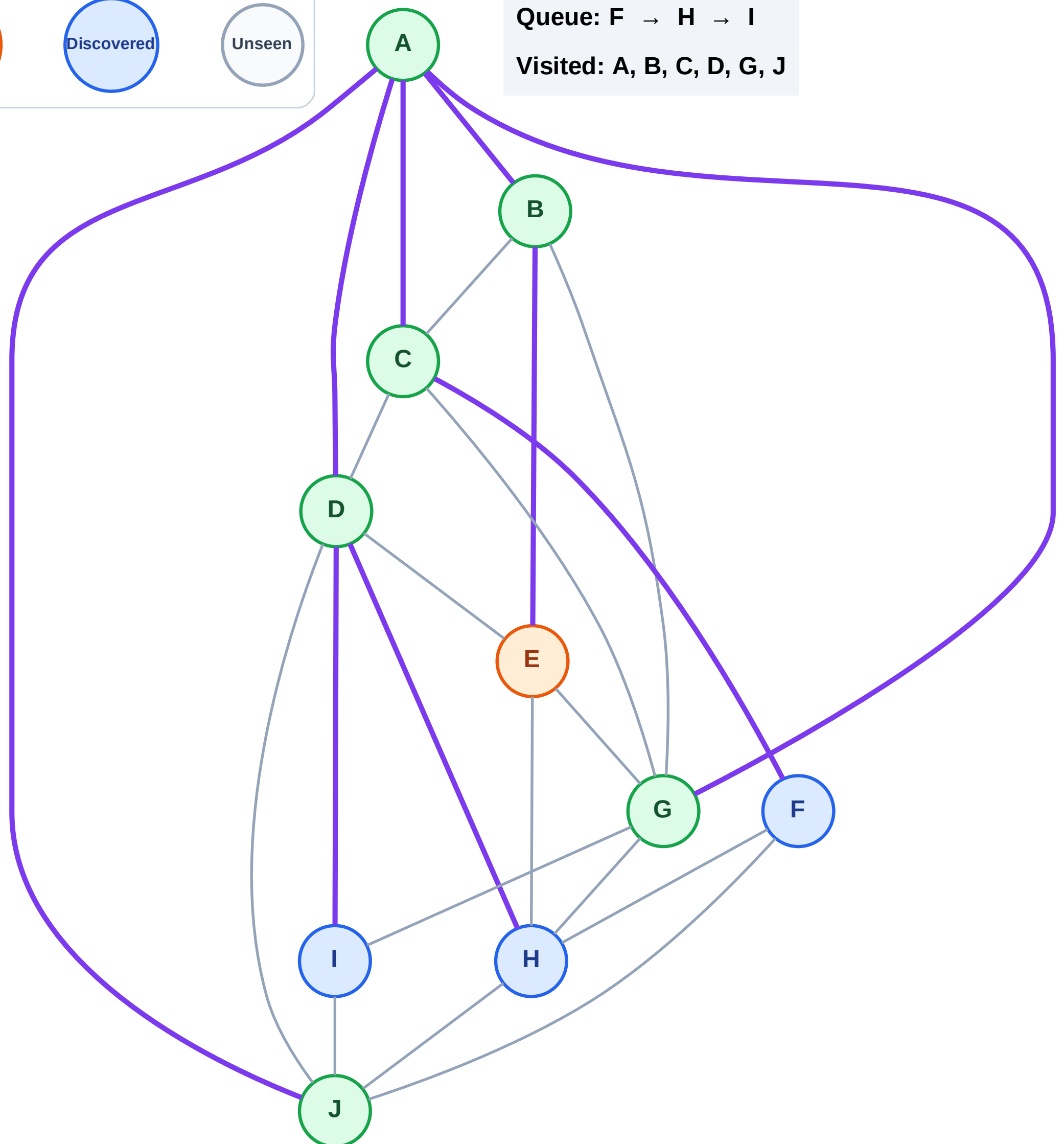
Step 14: Process node E

Legend



Queue: F → H → I

Visited: A, B, C, D, G, J



BFS Traversal

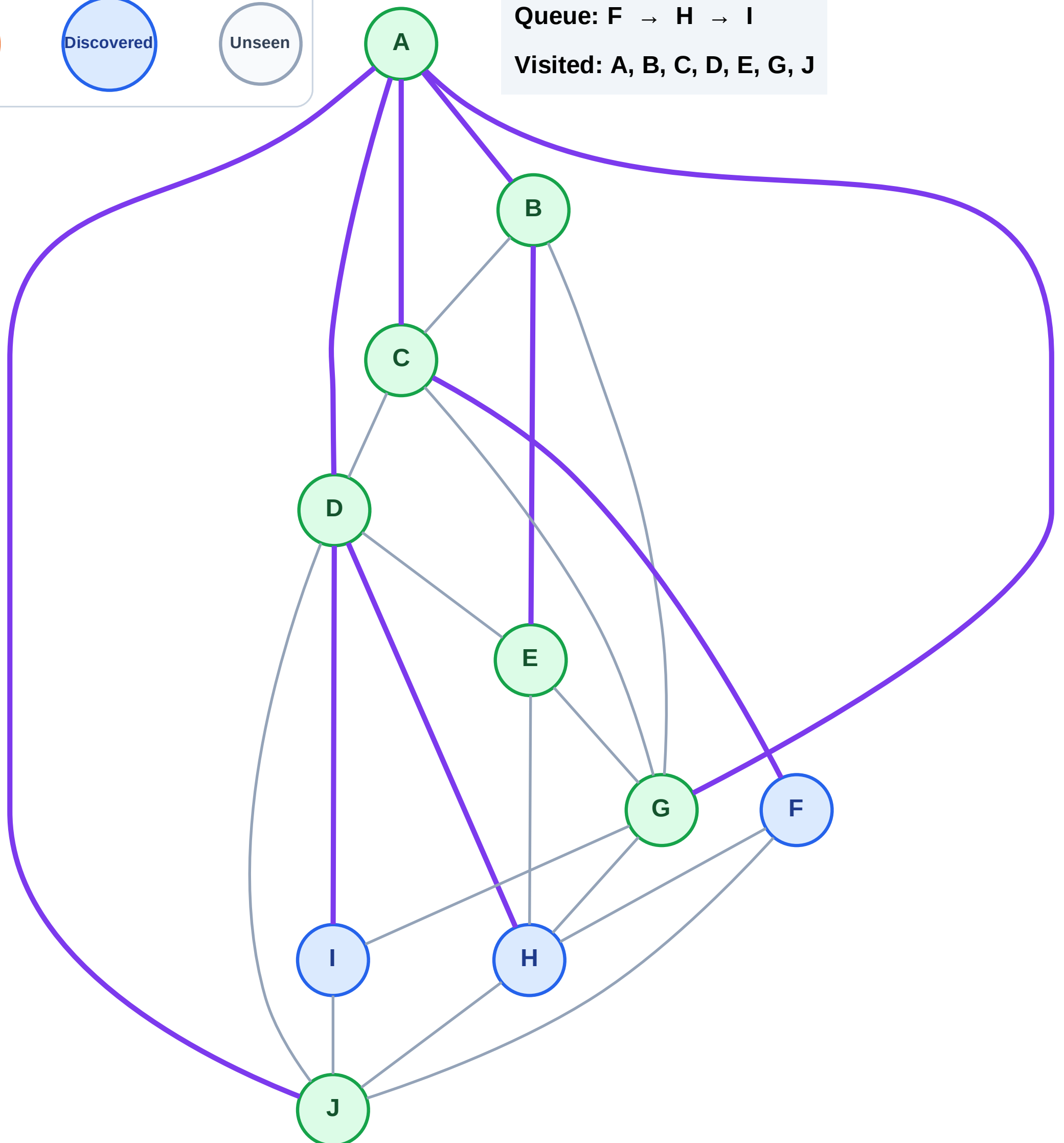
Step 15: No new neighbors from E

Legend



Queue: F → H → I

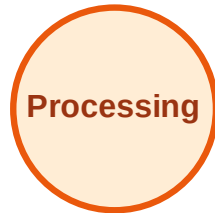
Visited: A, B, C, D, E, G, J



BFS Traversal

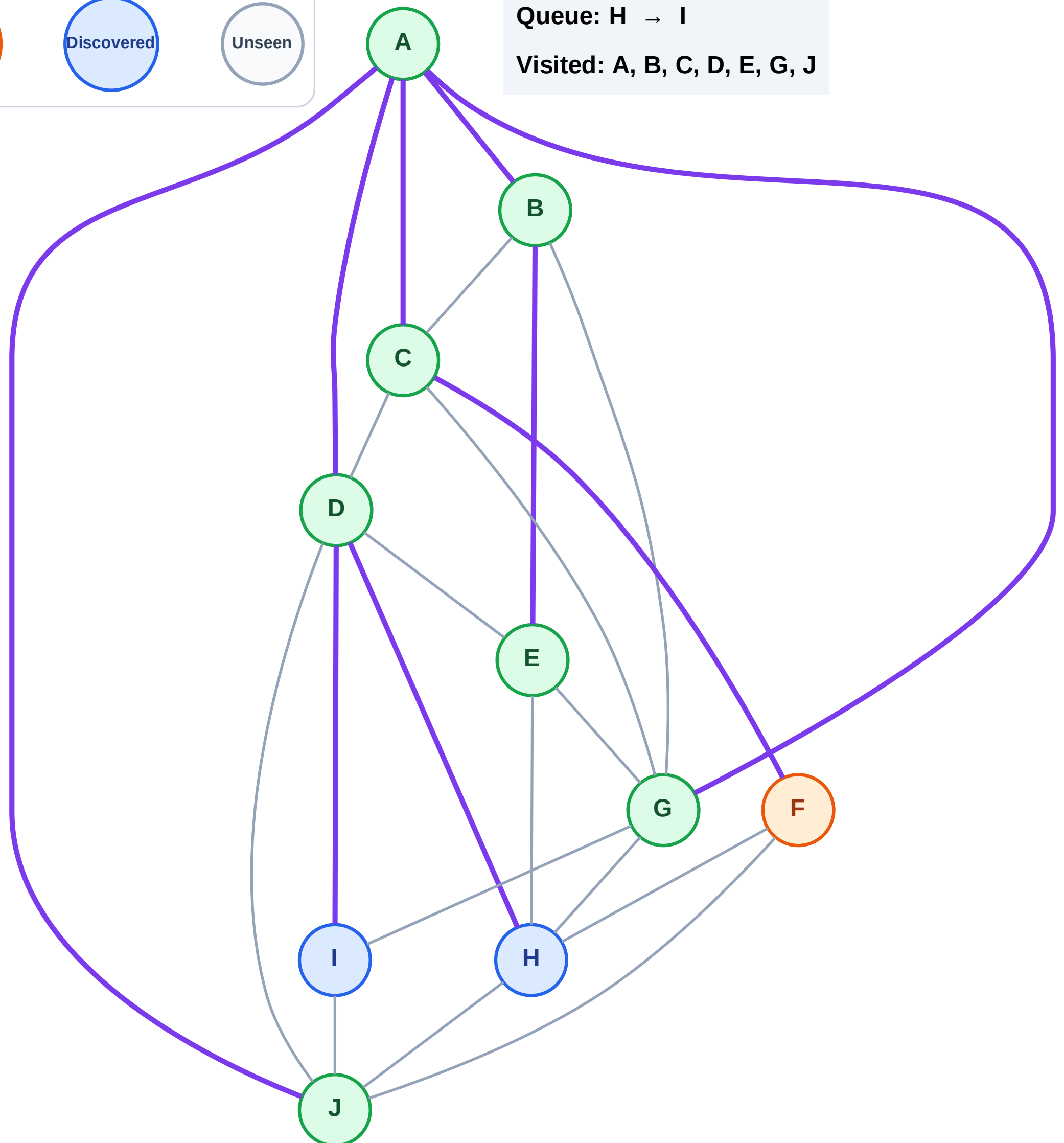
Step 16: Process node F

Legend



Queue: H → I

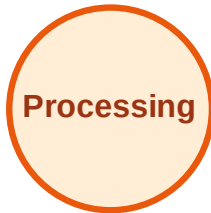
Visited: A, B, C, D, E, G, J



BFS Traversal

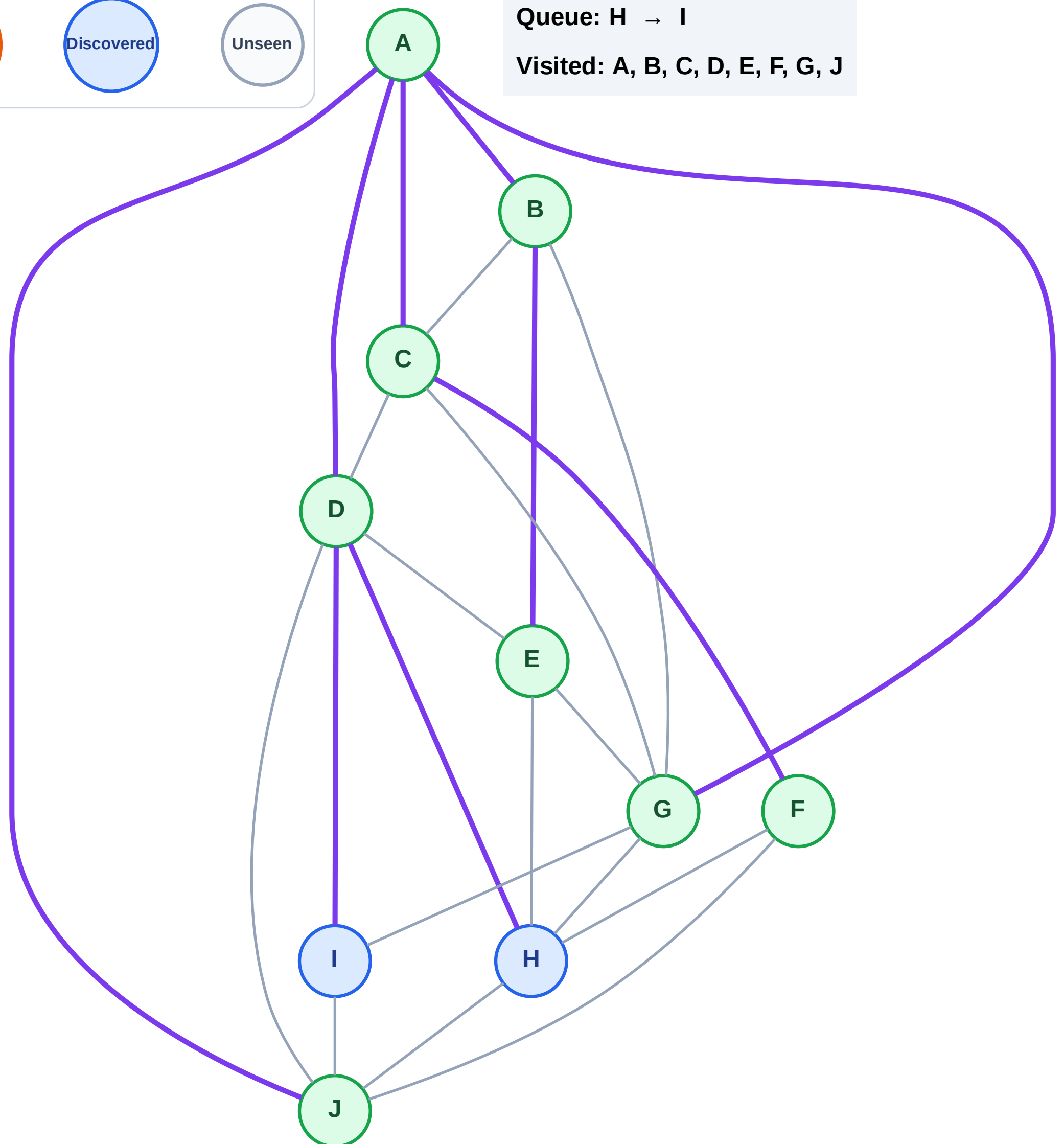
Step 17: No new neighbors from F

Legend



Queue: H → I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

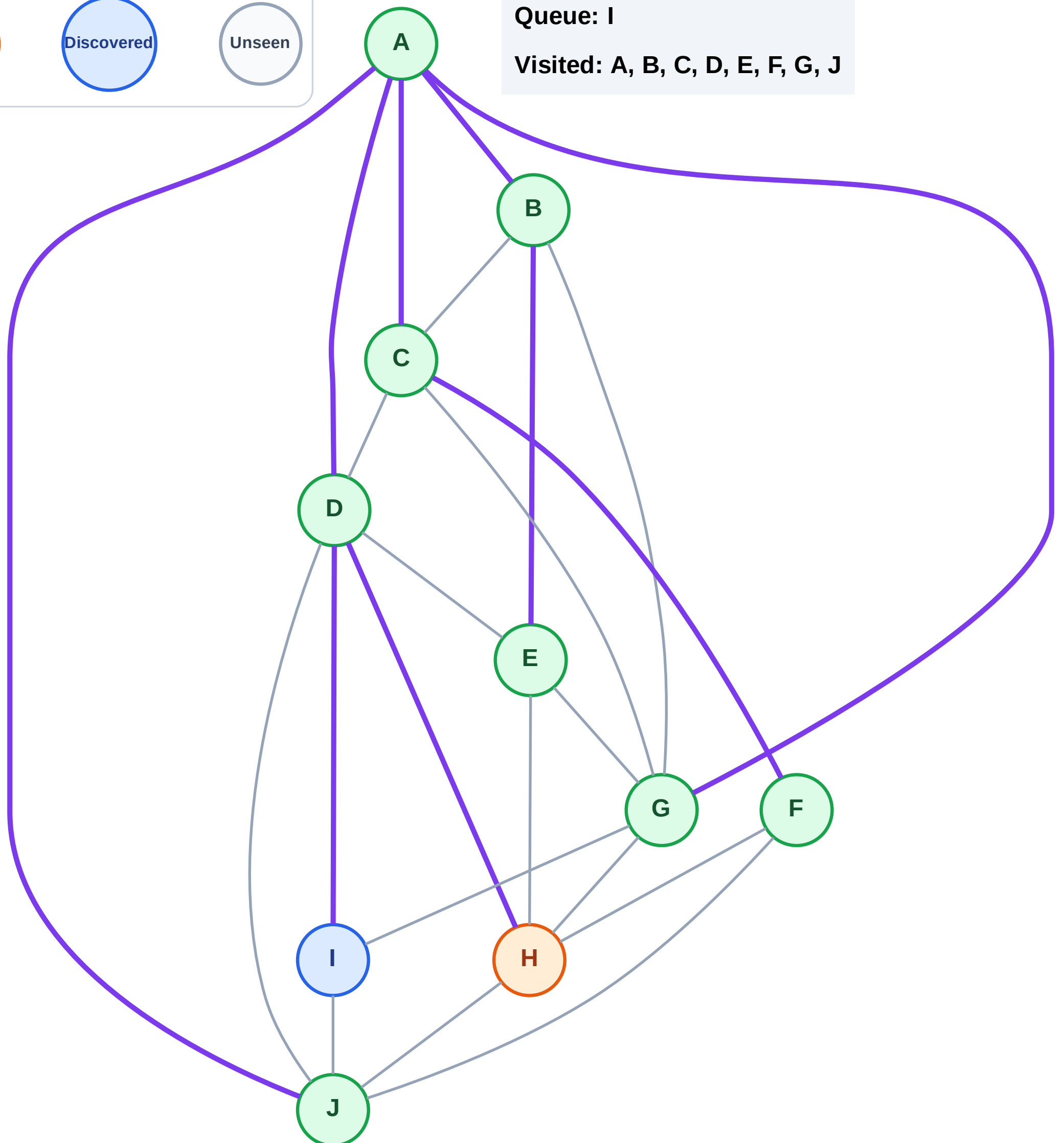
Step 18: Process node H

Legend



Queue: I

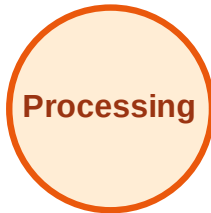
Visited: A, B, C, D, E, F, G, J



BFS Traversal

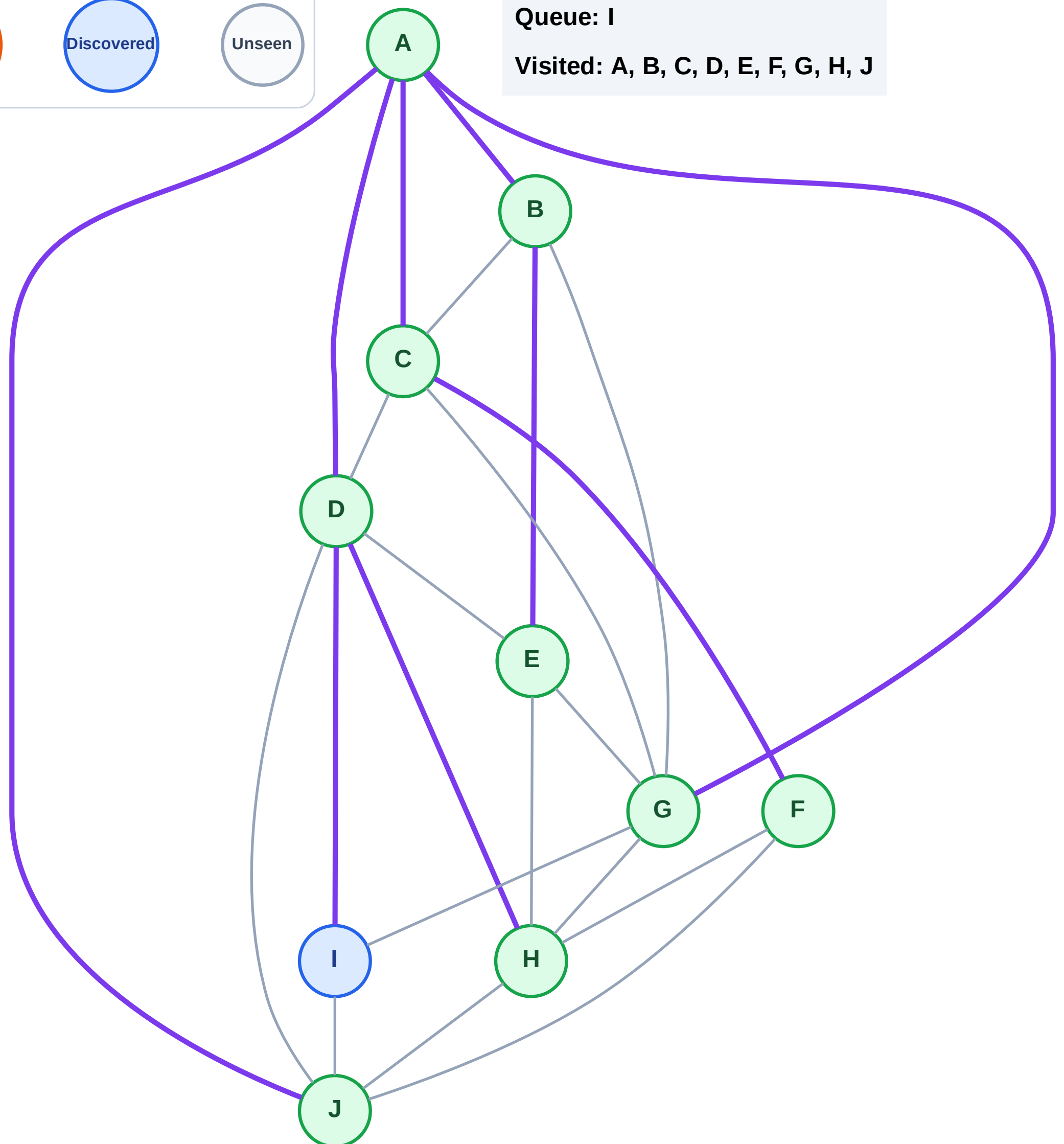
Step 19: No new neighbors from H

Legend



Queue: I

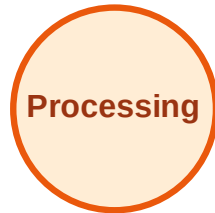
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

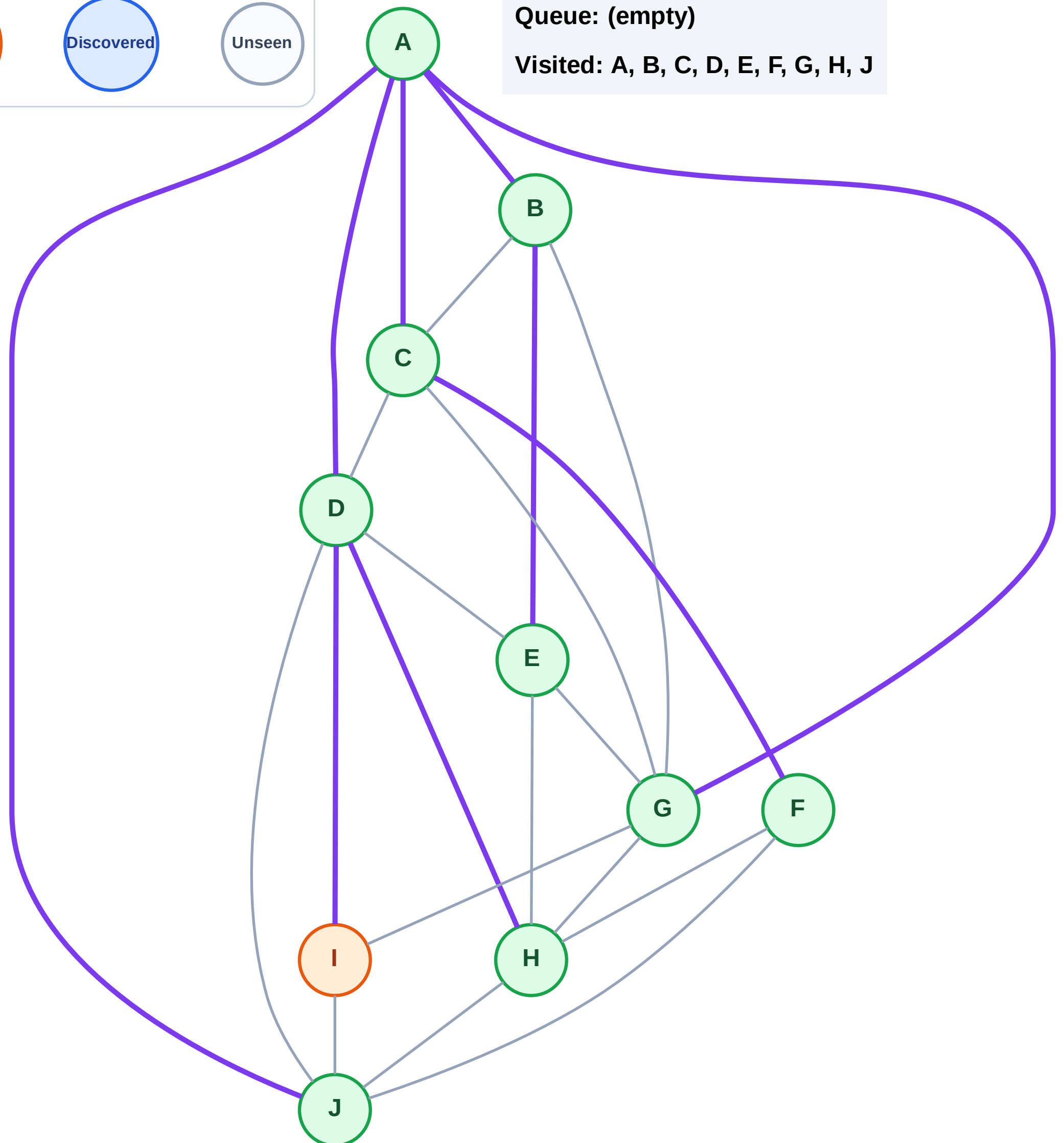
Step 20: Process node I

Legend



Queue: (empty)

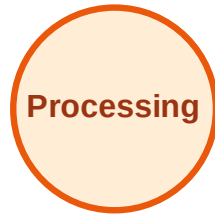
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

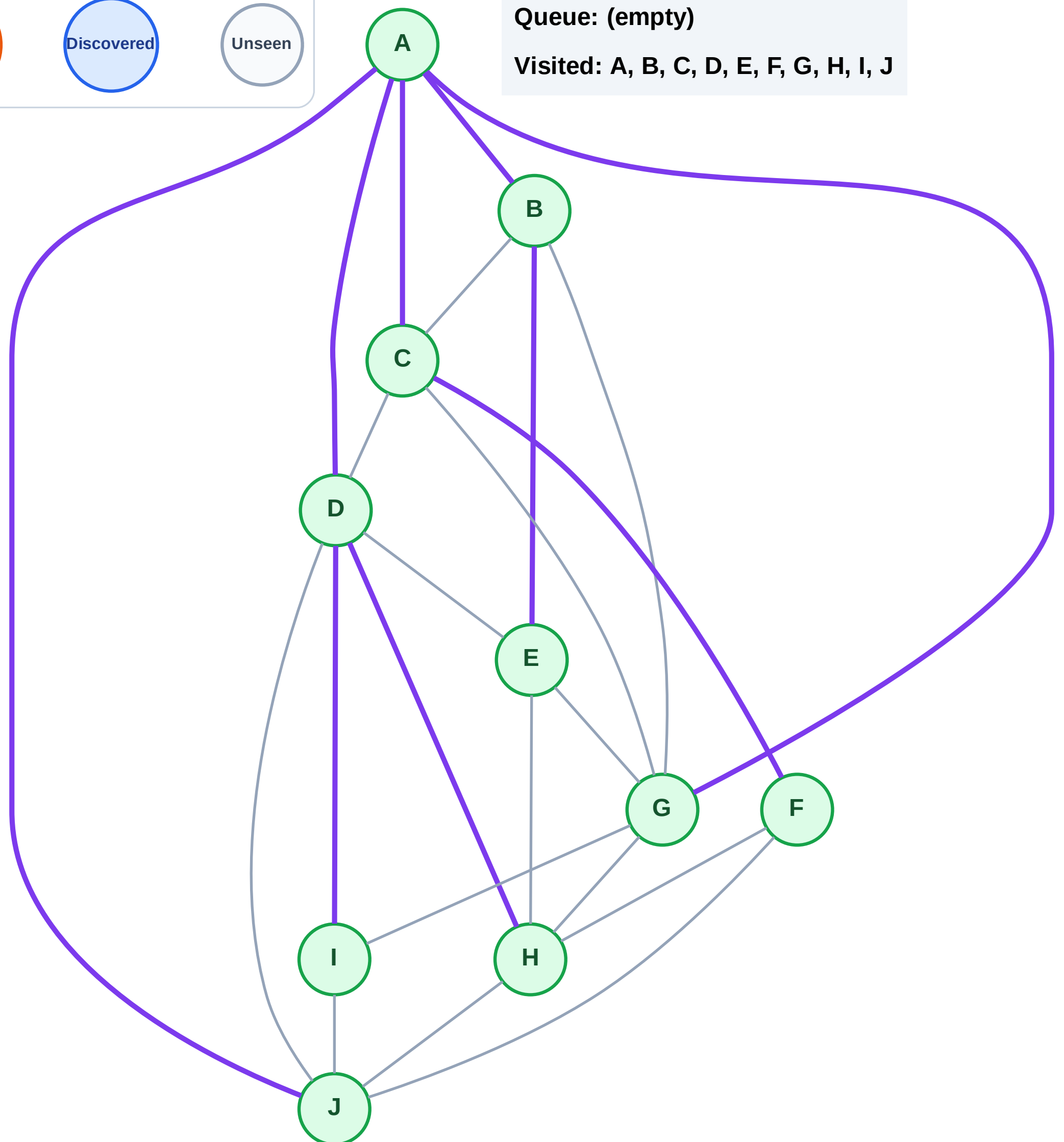
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

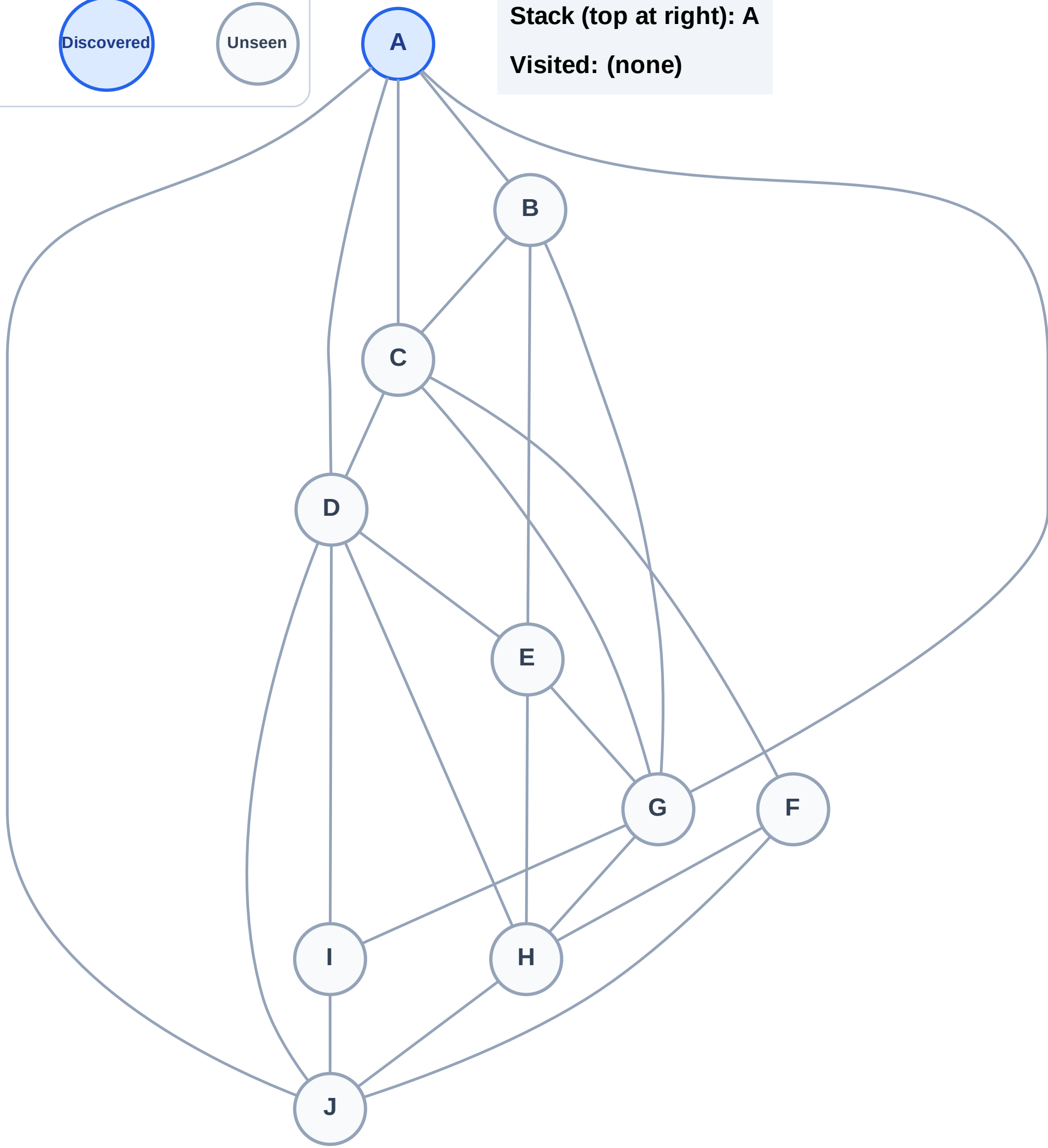
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

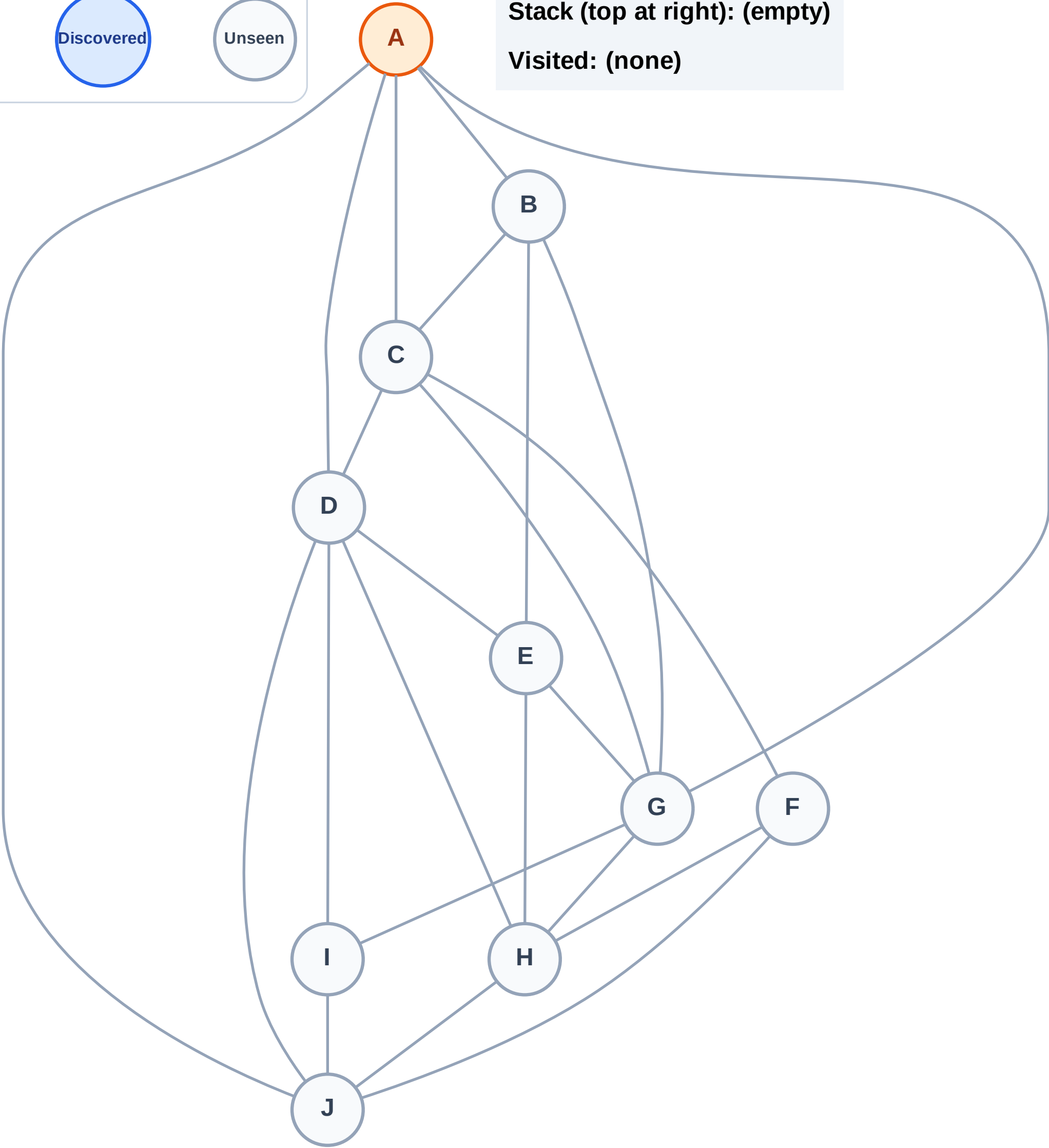
Complete

Processing

Discovered

Unseen

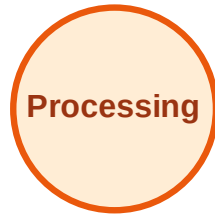
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

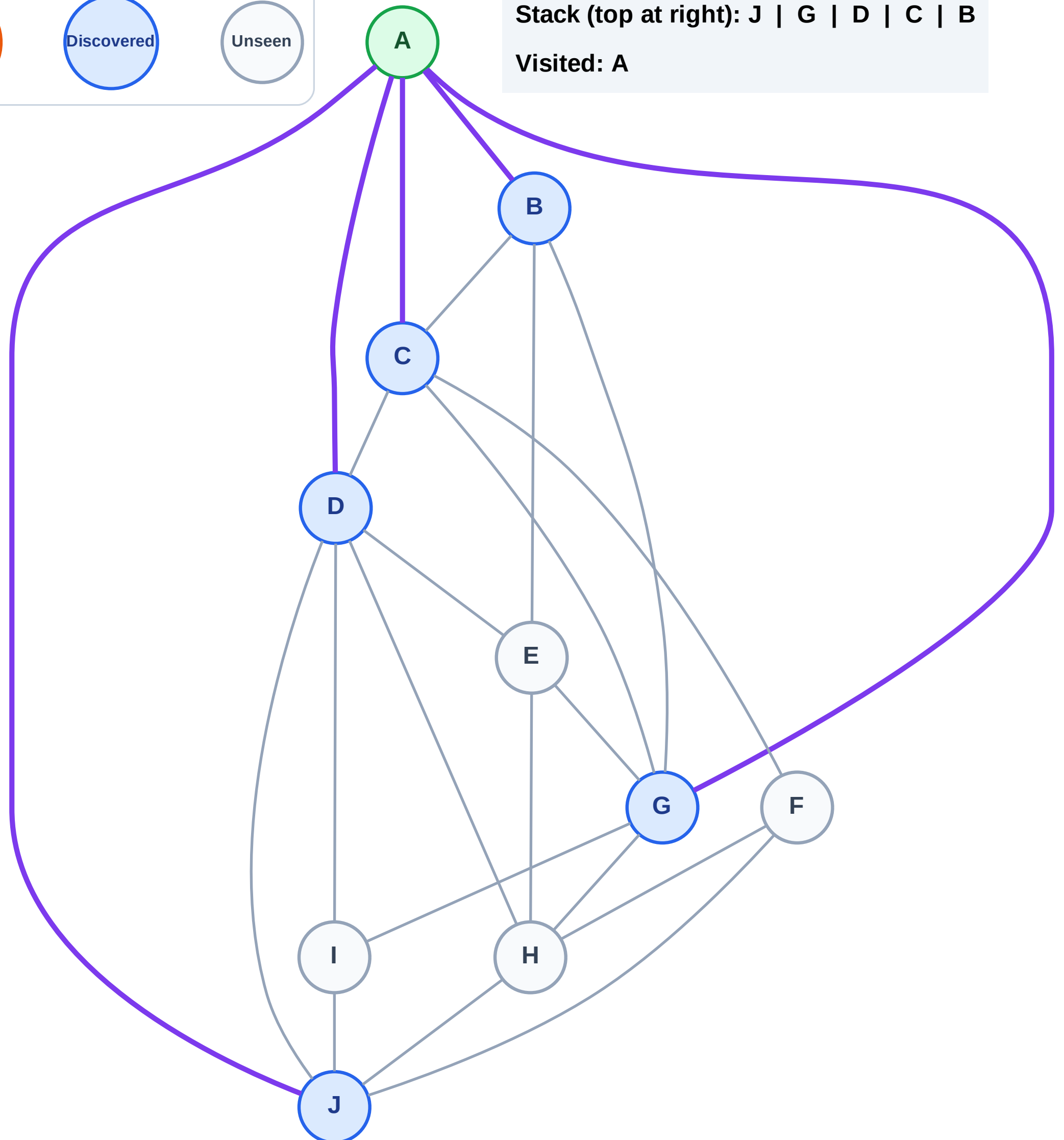
Step 3: Discover J, G, D, C, B from A

Legend



Stack (top at right): J | G | D | C | B

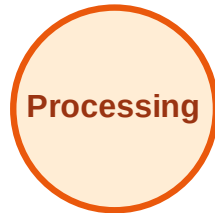
Visited: A



DFS Traversal

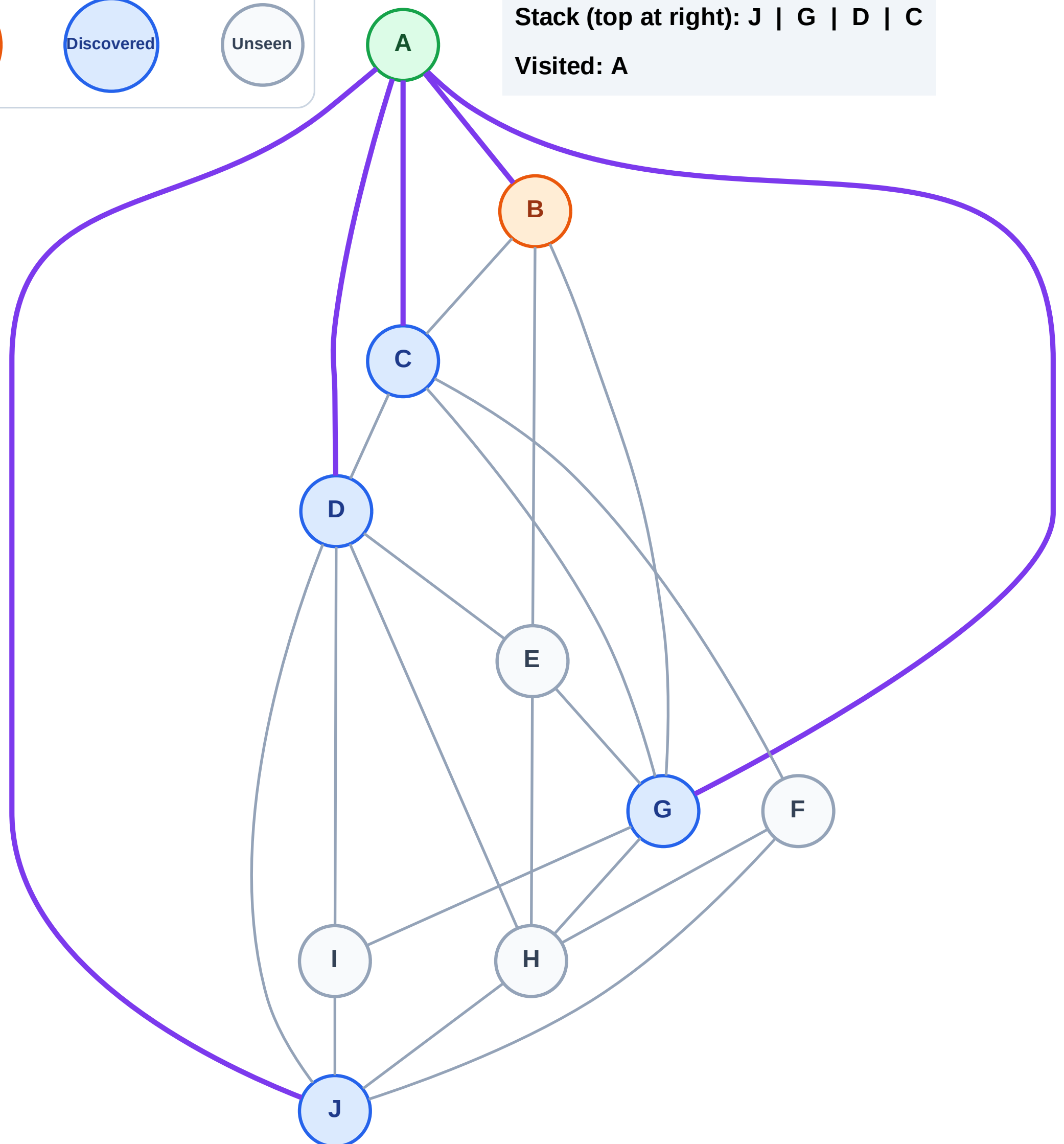
Step 4: Process node B

Legend



Stack (top at right): J | G | D | C

Visited: A



DFS Traversal

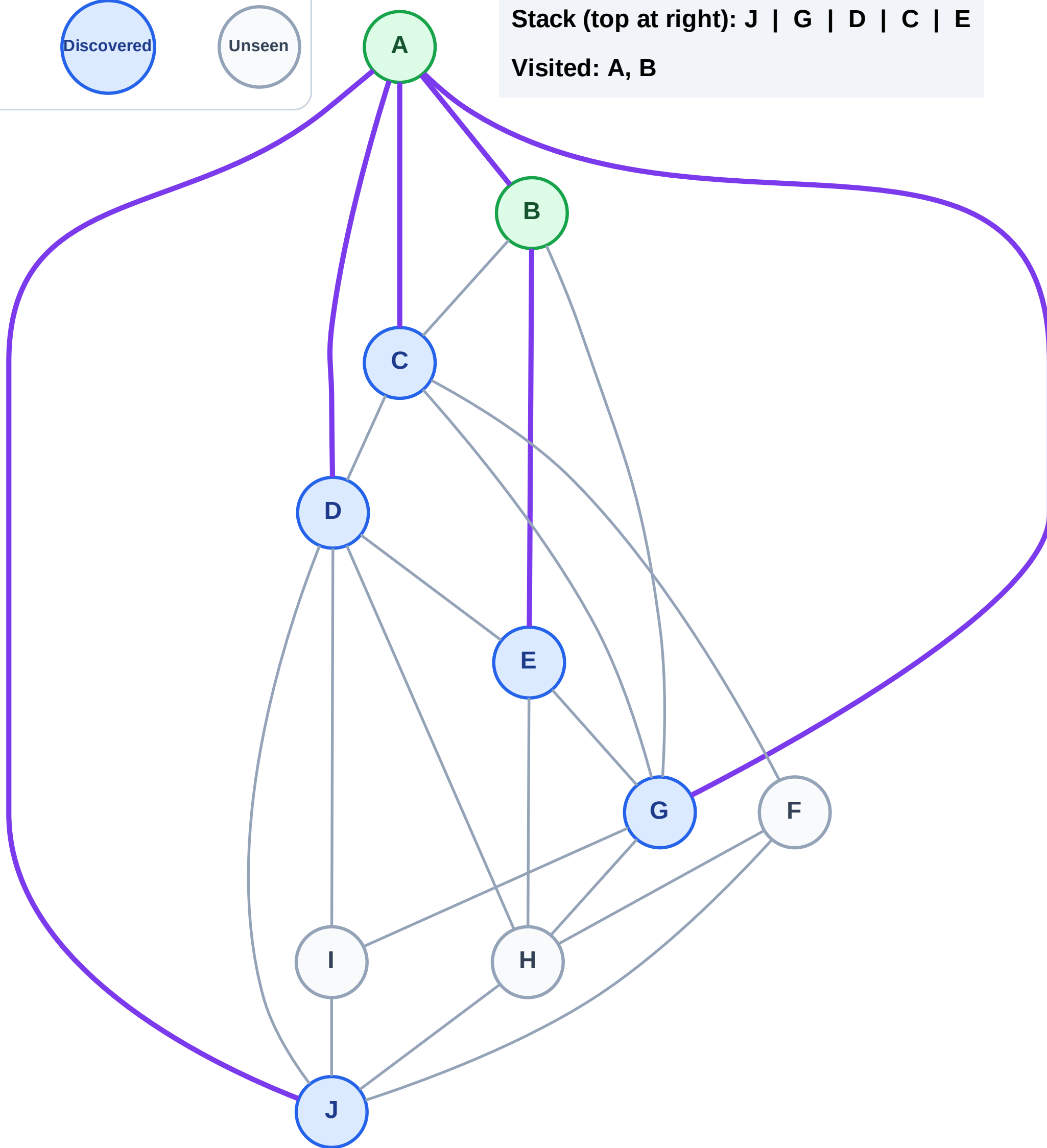
Step 5: Discover E from B

Legend



Stack (top at right): J | G | D | C | E

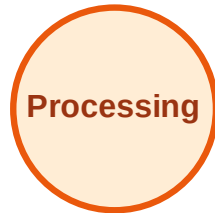
Visited: A, B



DFS Traversal

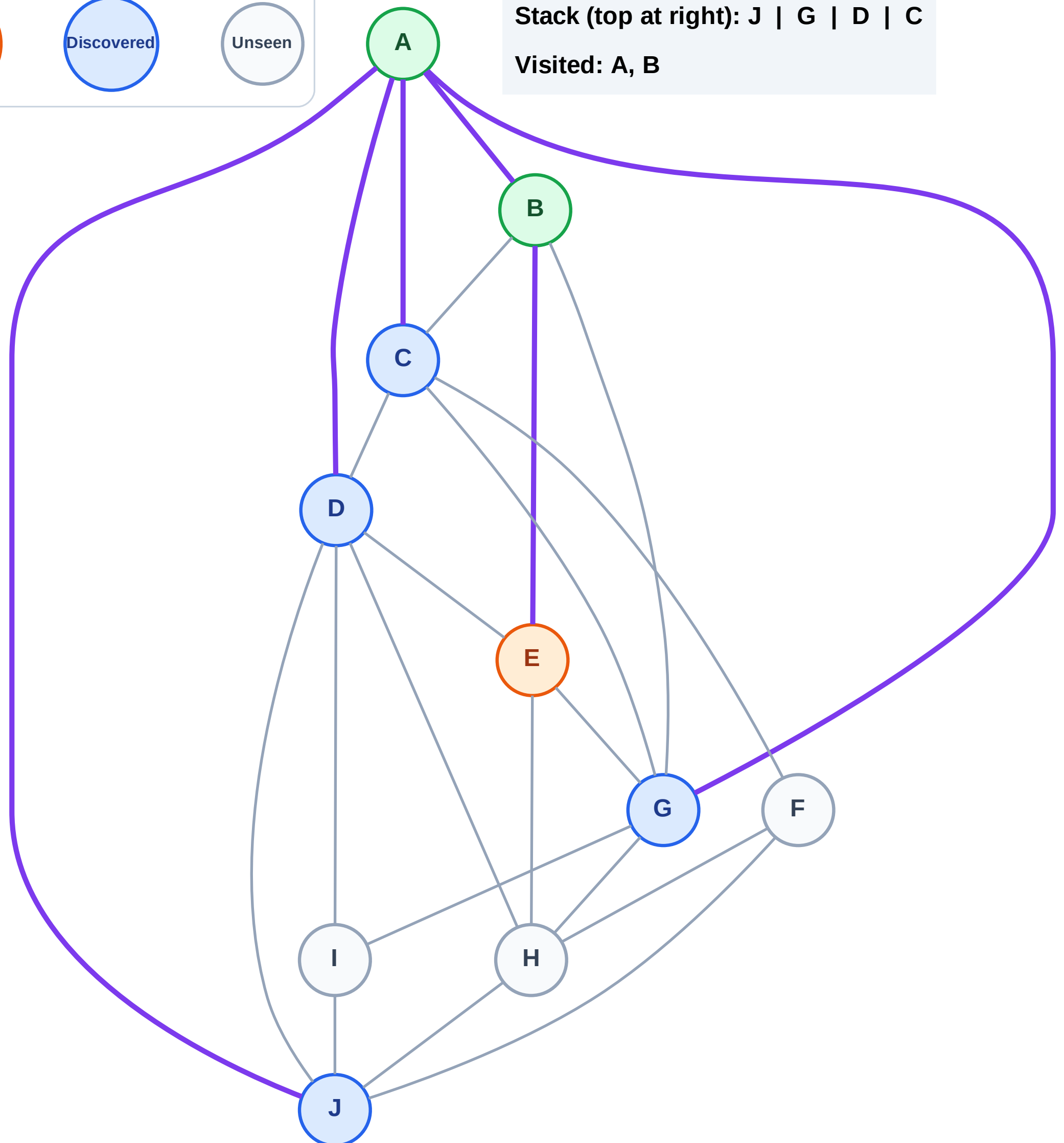
Step 6: Process node E

Legend



Stack (top at right): J | G | D | C

Visited: A, B



DFS Traversal

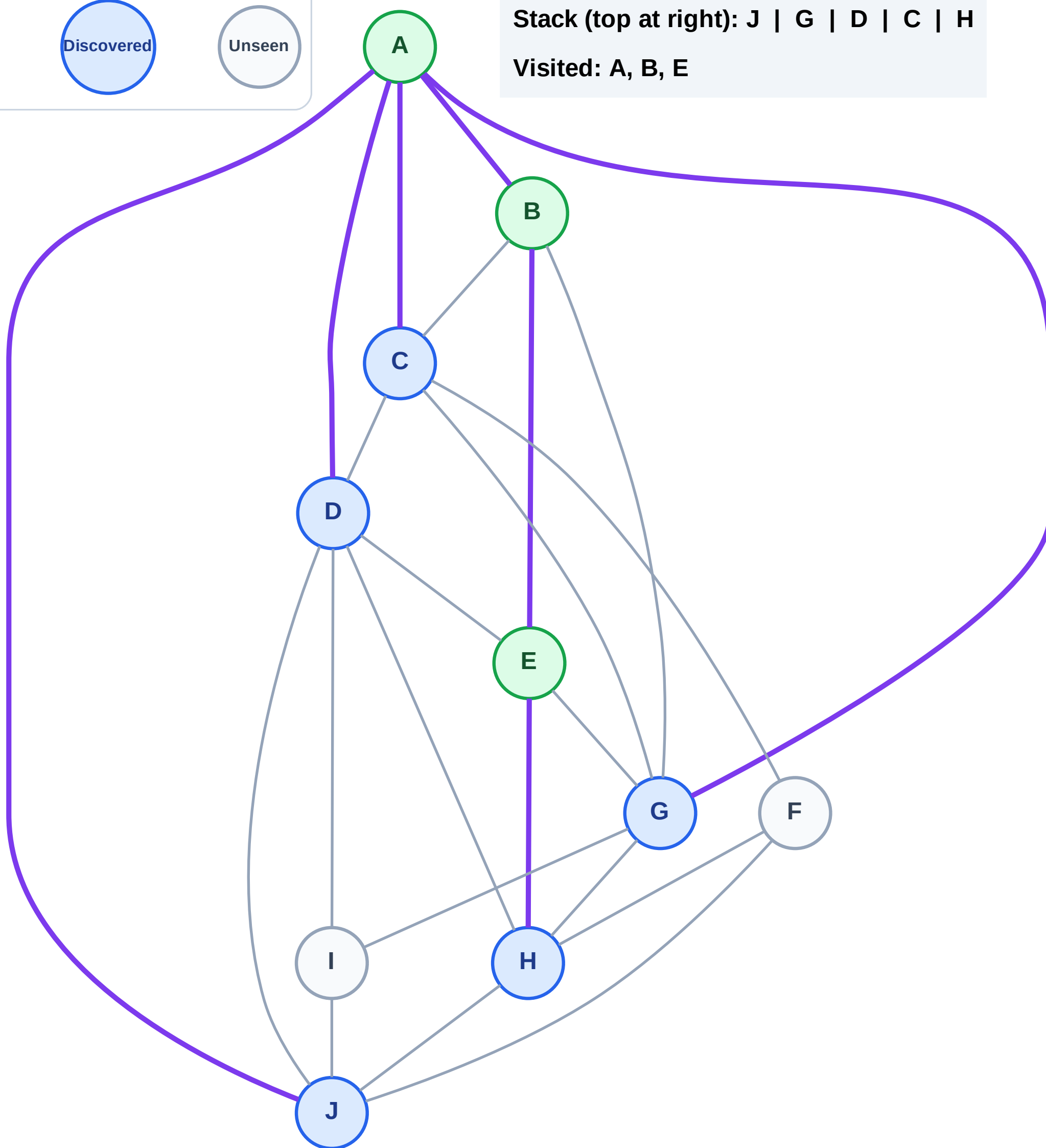
Step 7: Discover H from E

Legend



Stack (top at right): J | G | D | C | H

Visited: A, B, E



DFS Traversal

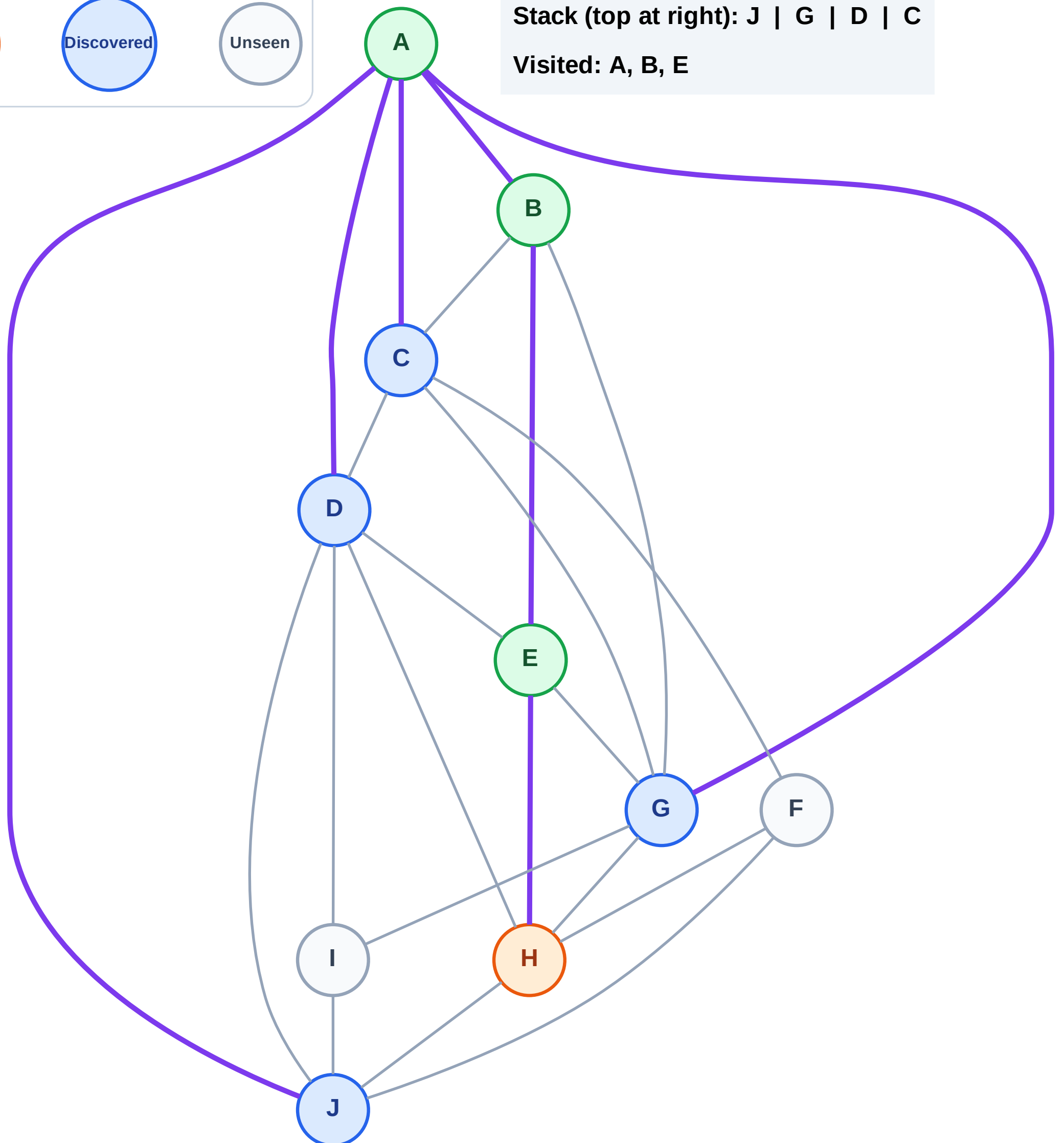
Step 8: Process node H

Legend



Stack (top at right): J | G | D | C

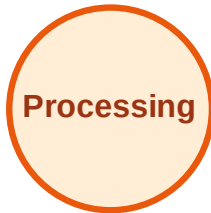
Visited: A, B, E



DFS Traversal

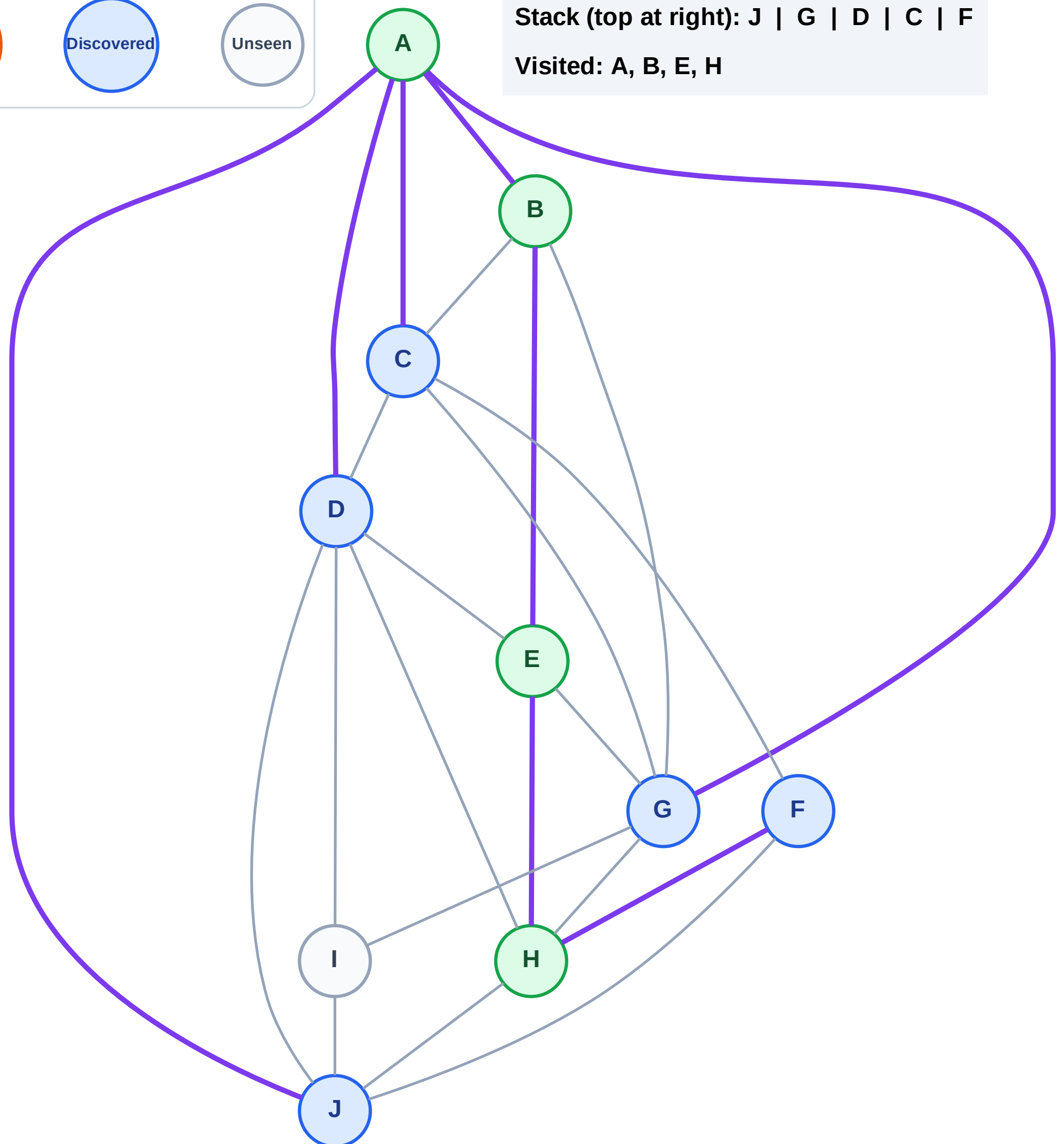
Step 9: Discover F from H

Legend



Stack (top at right): J | G | D | C | F

Visited: A, B, E, H



DFS Traversal

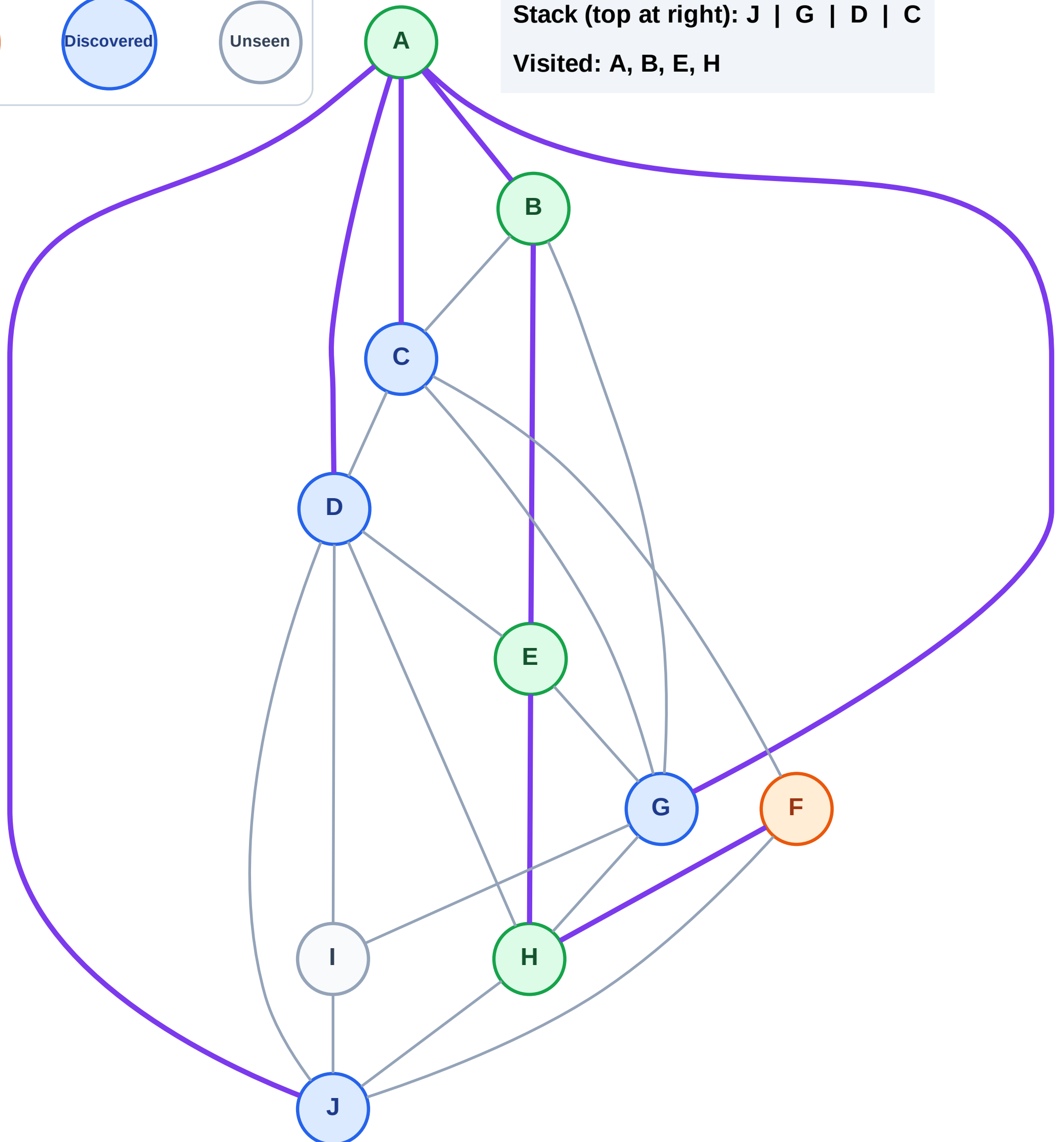
Step 10: Process node F

Legend



Stack (top at right): J | G | D | C

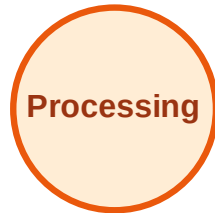
Visited: A, B, E, H



DFS Traversal

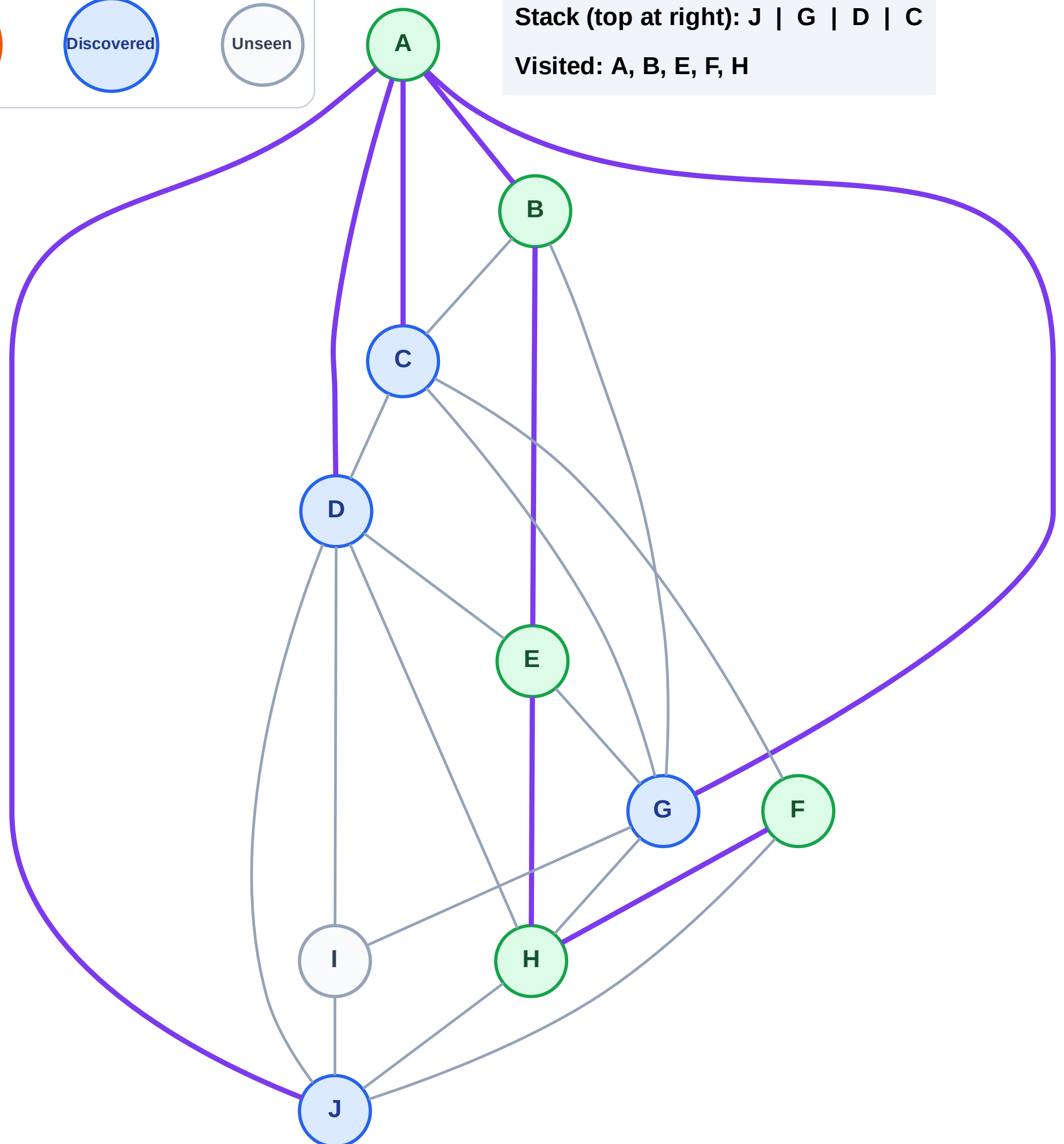
Step 11: No new neighbors from F

Legend



Stack (top at right): J | G | D | C

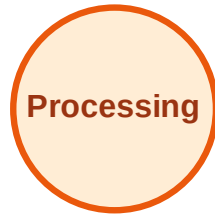
Visited: A, B, E, F, H



DFS Traversal

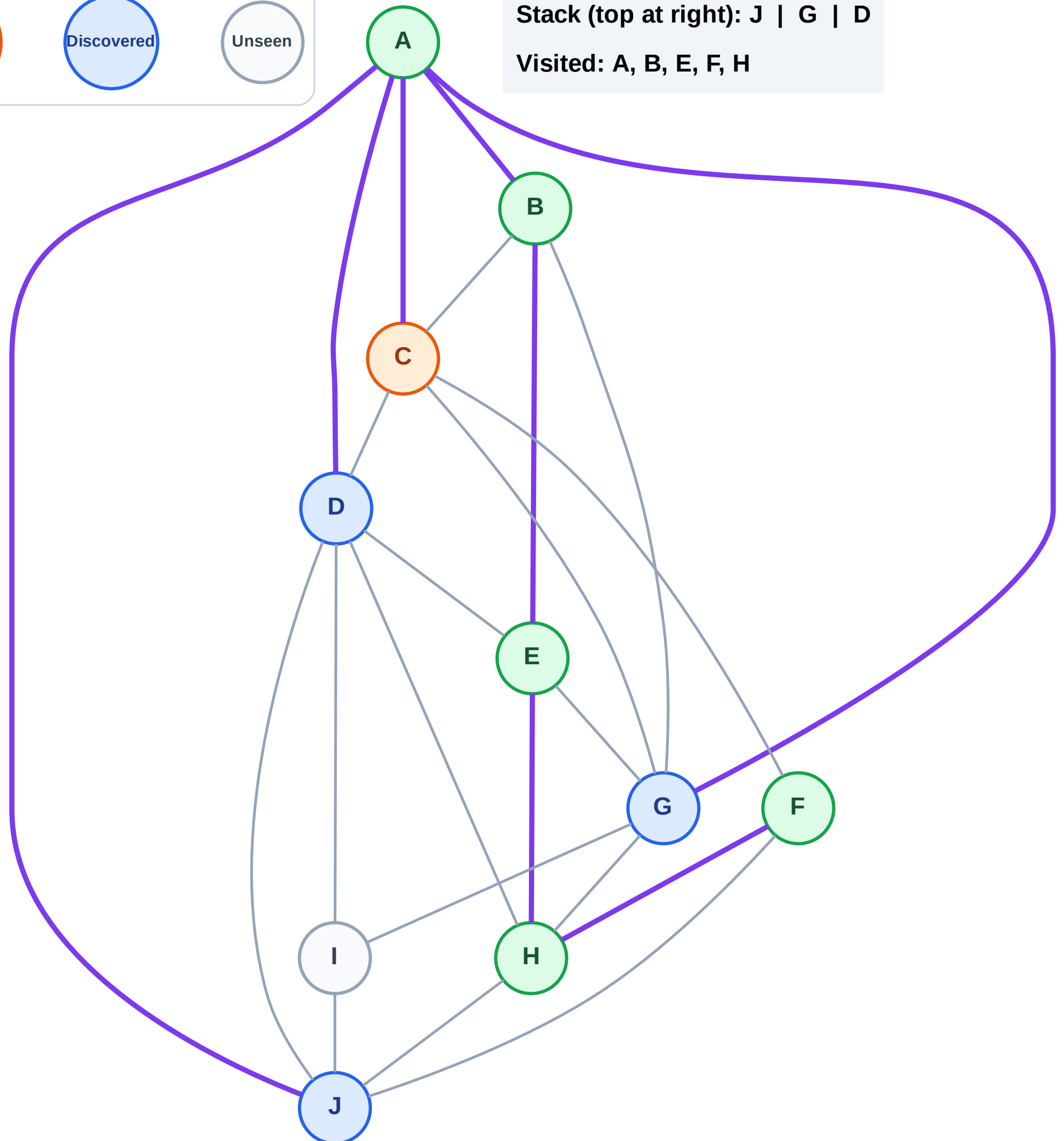
Step 12: Process node C

Legend



Stack (top at right): J | G | D

Visited: A, B, E, F, H



DFS Traversal

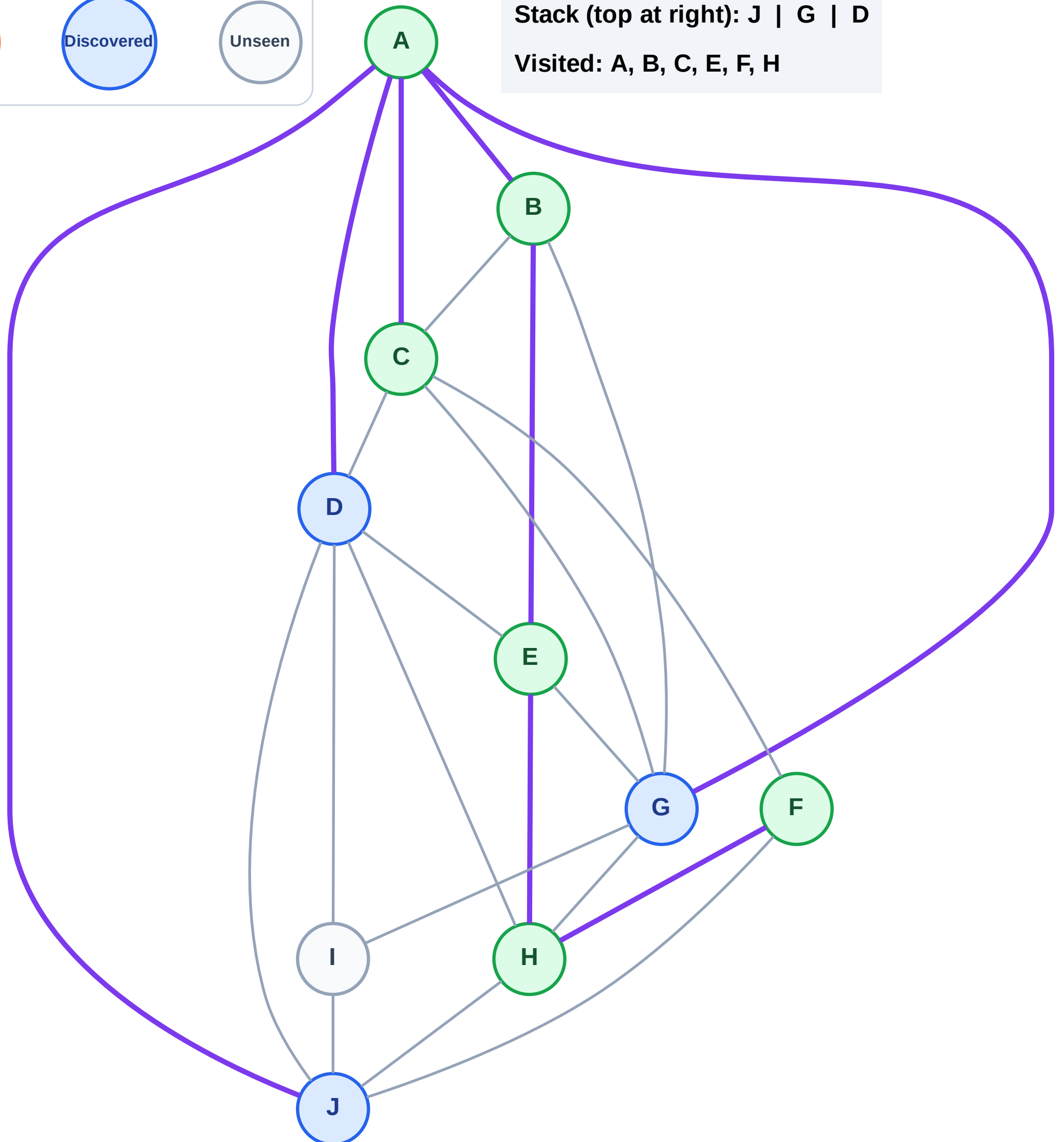
Step 13: No new neighbors from C

Legend



Stack (top at right): J | G | D

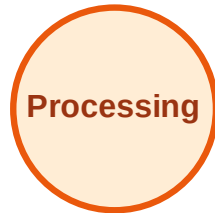
Visited: A, B, C, E, F, H



DFS Traversal

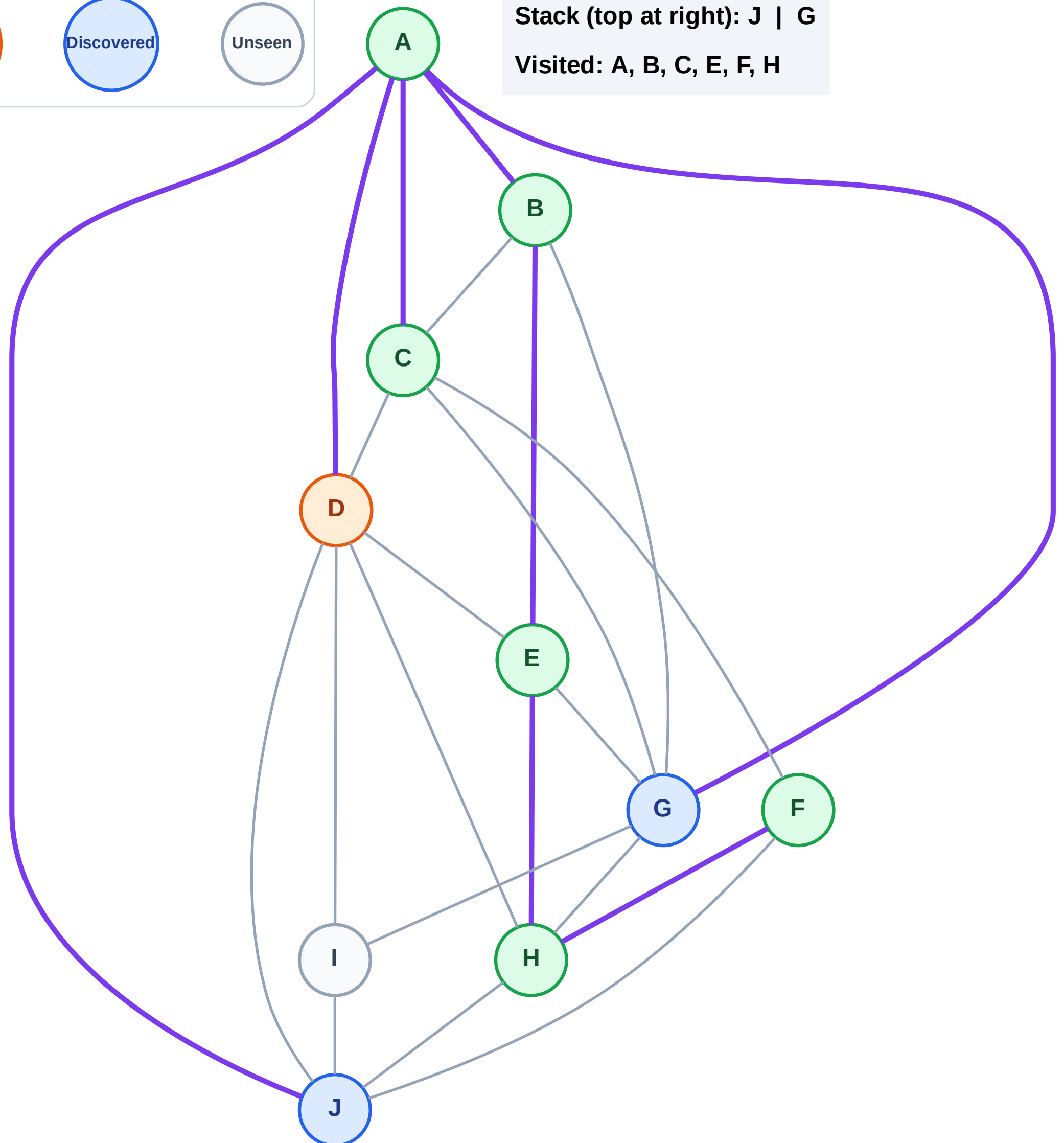
Step 14: Process node D

Legend



Stack (top at right): J | G

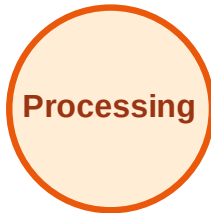
Visited: A, B, C, E, F, H



DFS Traversal

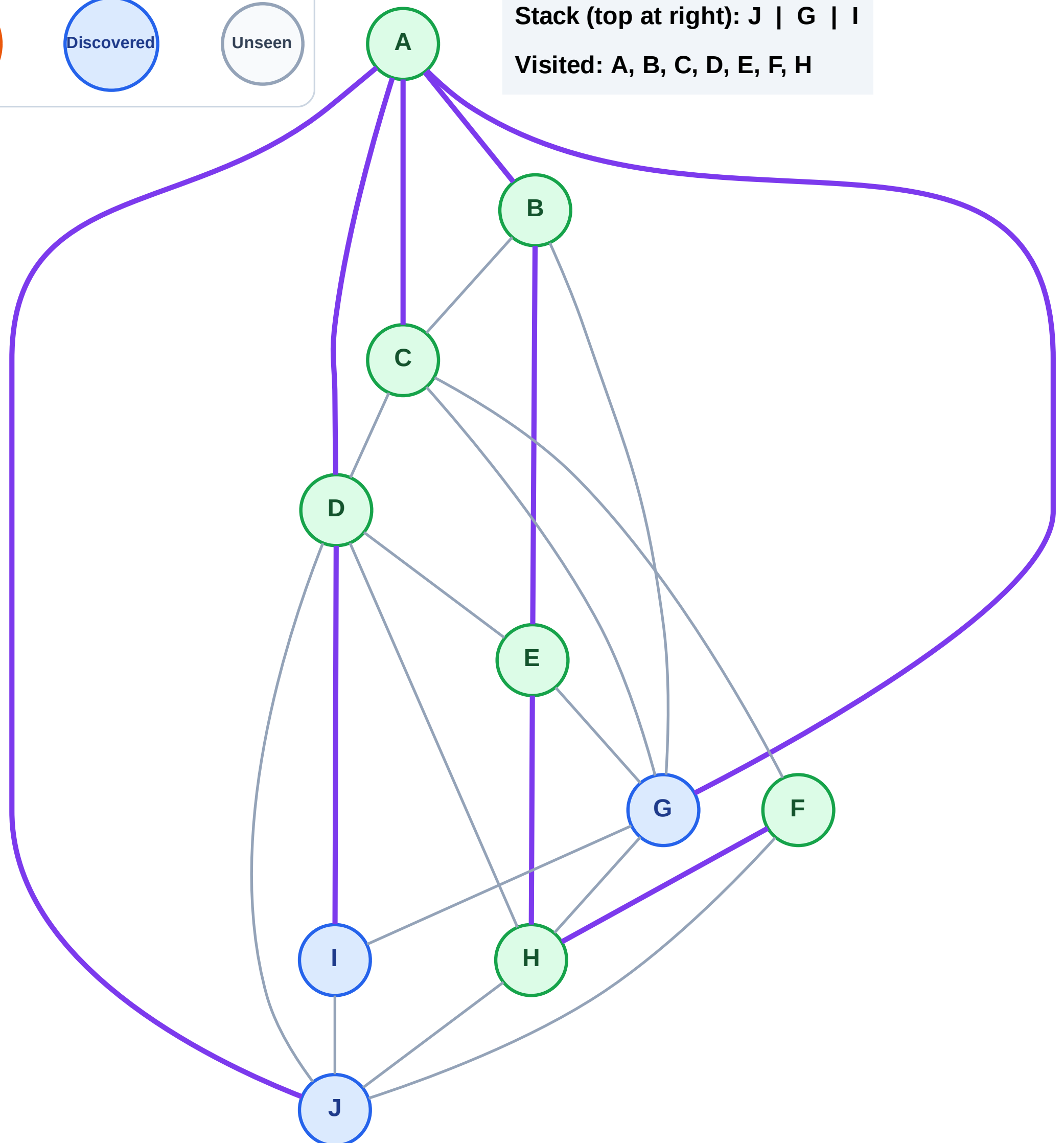
Step 15: Discover I from D

Legend



Stack (top at right): J | G | I

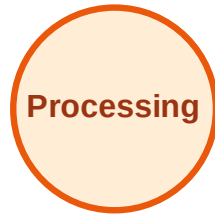
Visited: A, B, C, D, E, F, H



DFS Traversal

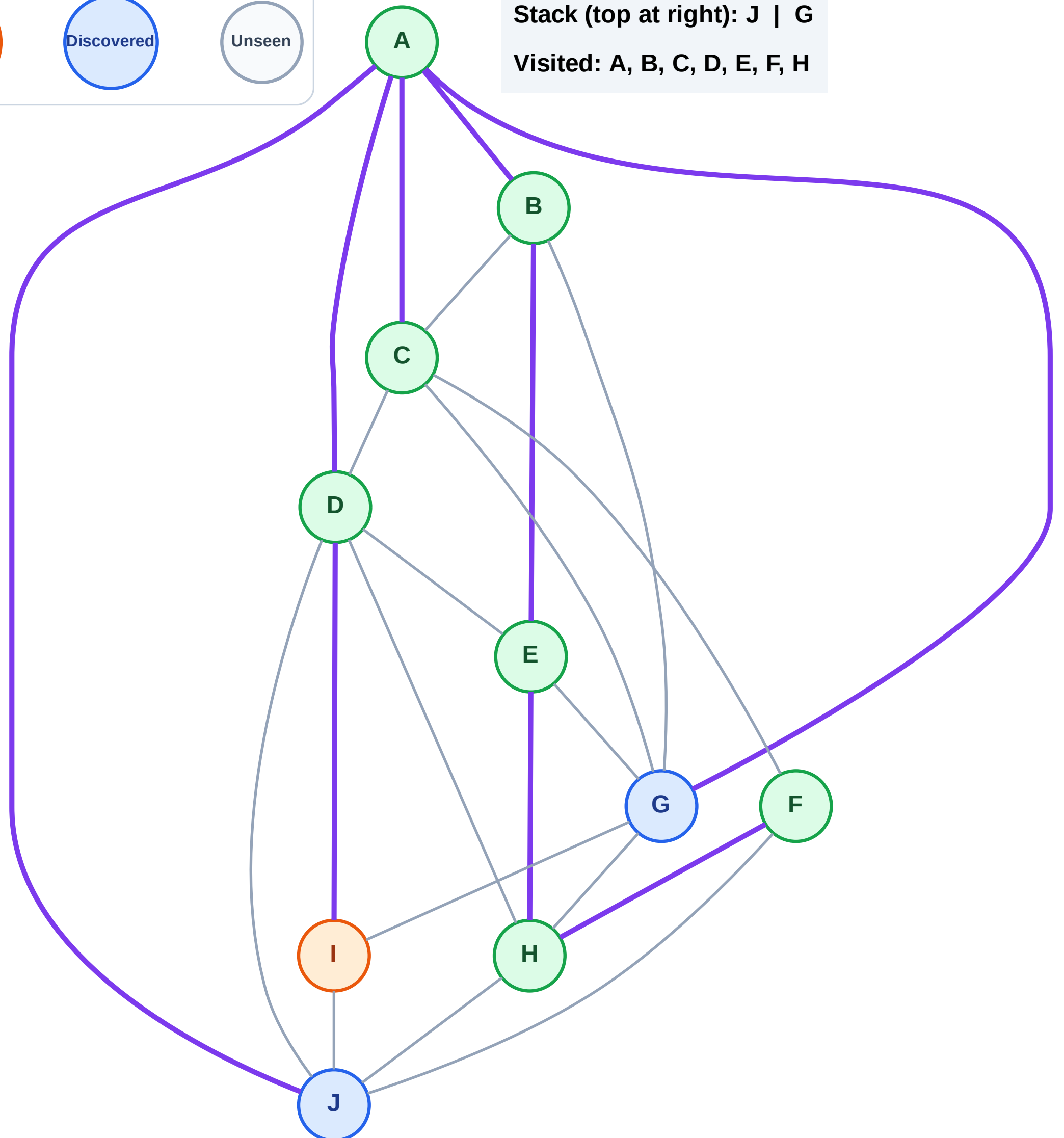
Step 16: Process node I

Legend



Stack (top at right): J | G

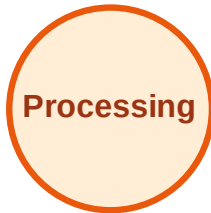
Visited: A, B, C, D, E, F, H



DFS Traversal

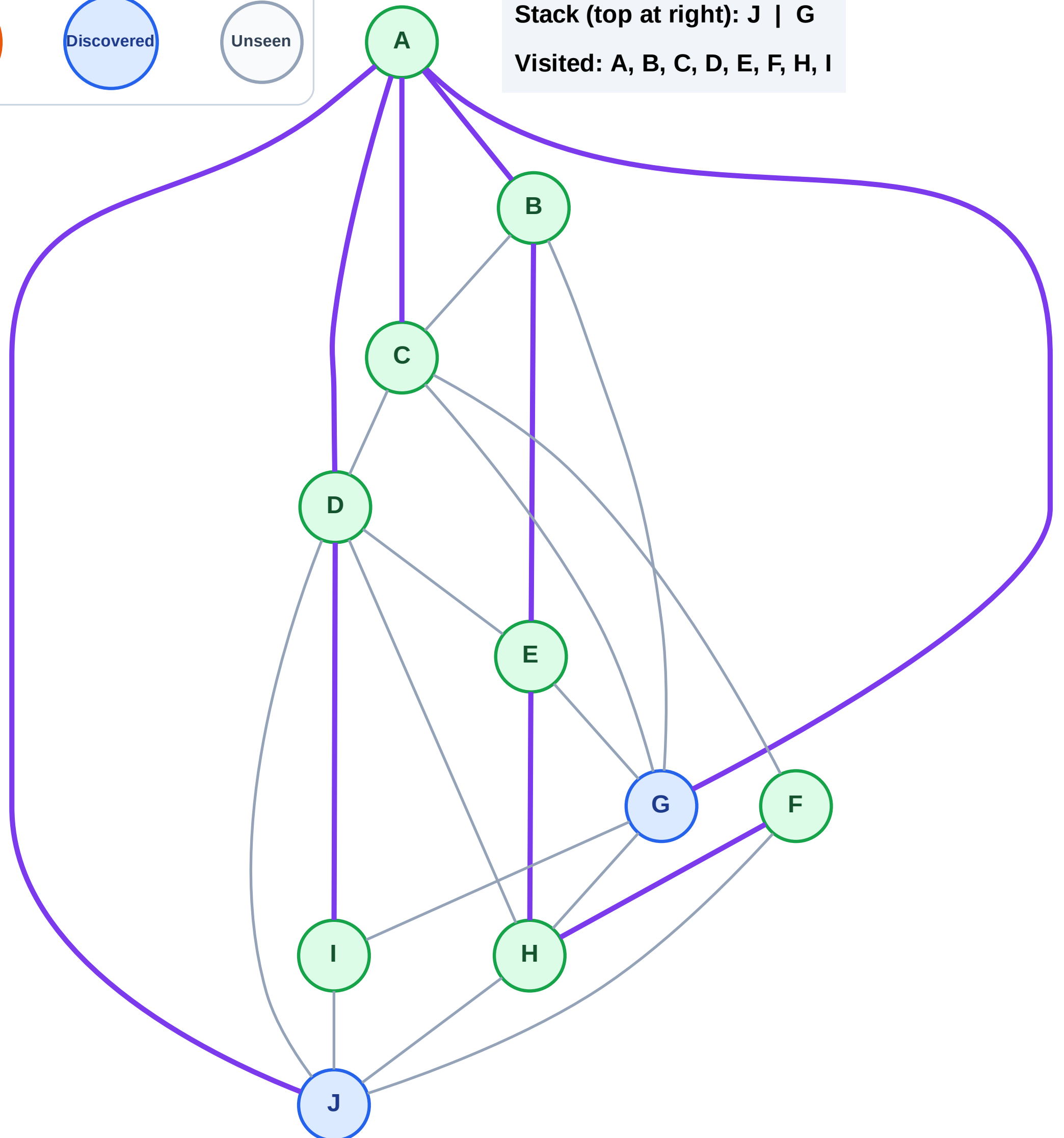
Step 17: No new neighbors from I

Legend



Stack (top at right): J | G

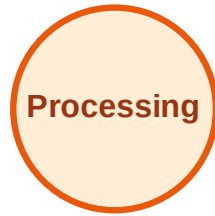
Visited: A, B, C, D, E, F, H, I



DFS Traversal

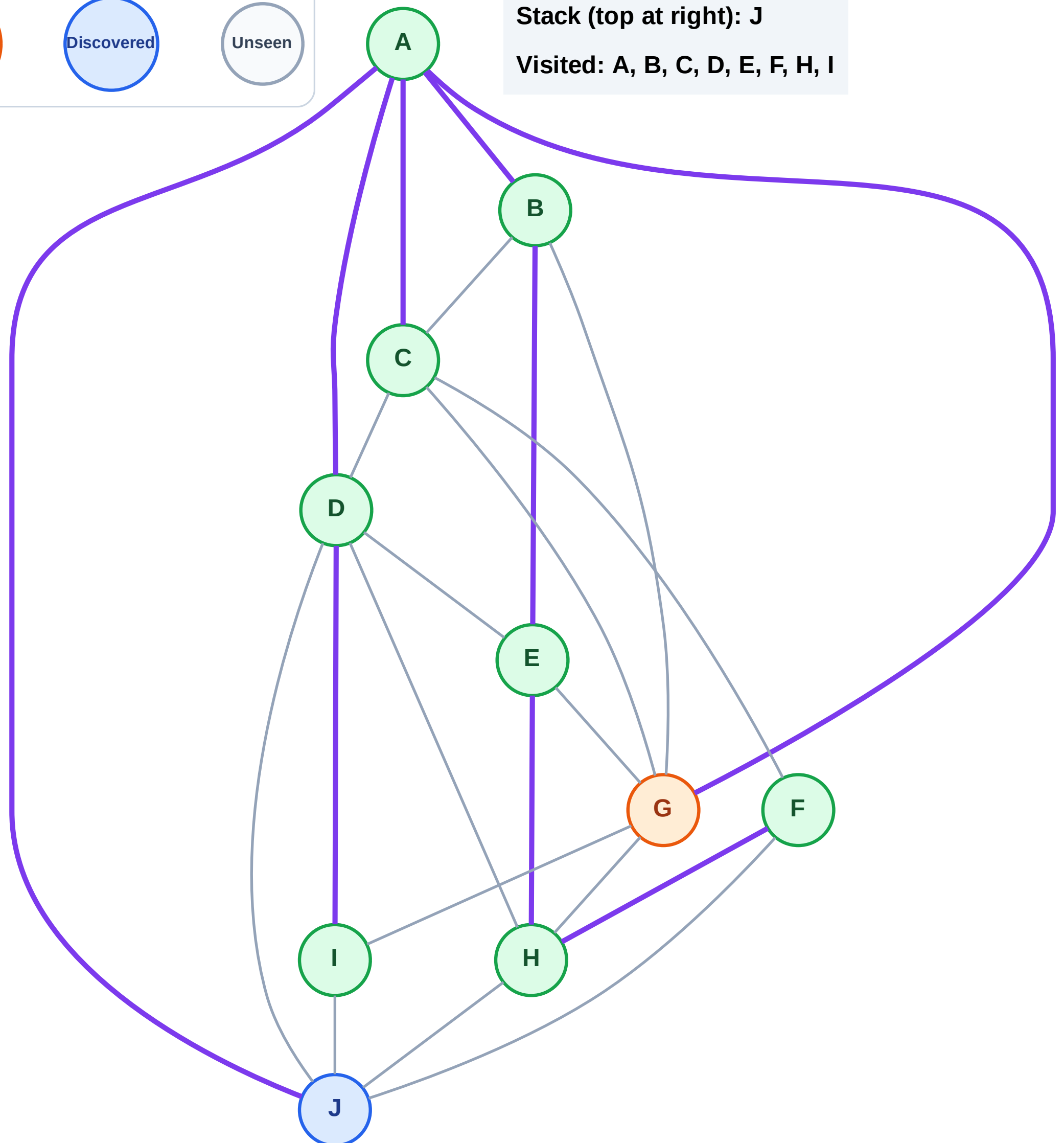
Step 18: Process node G

Legend



Stack (top at right): J

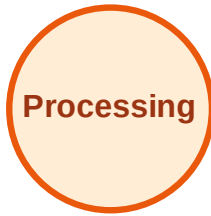
Visited: A, B, C, D, E, F, H, I



DFS Traversal

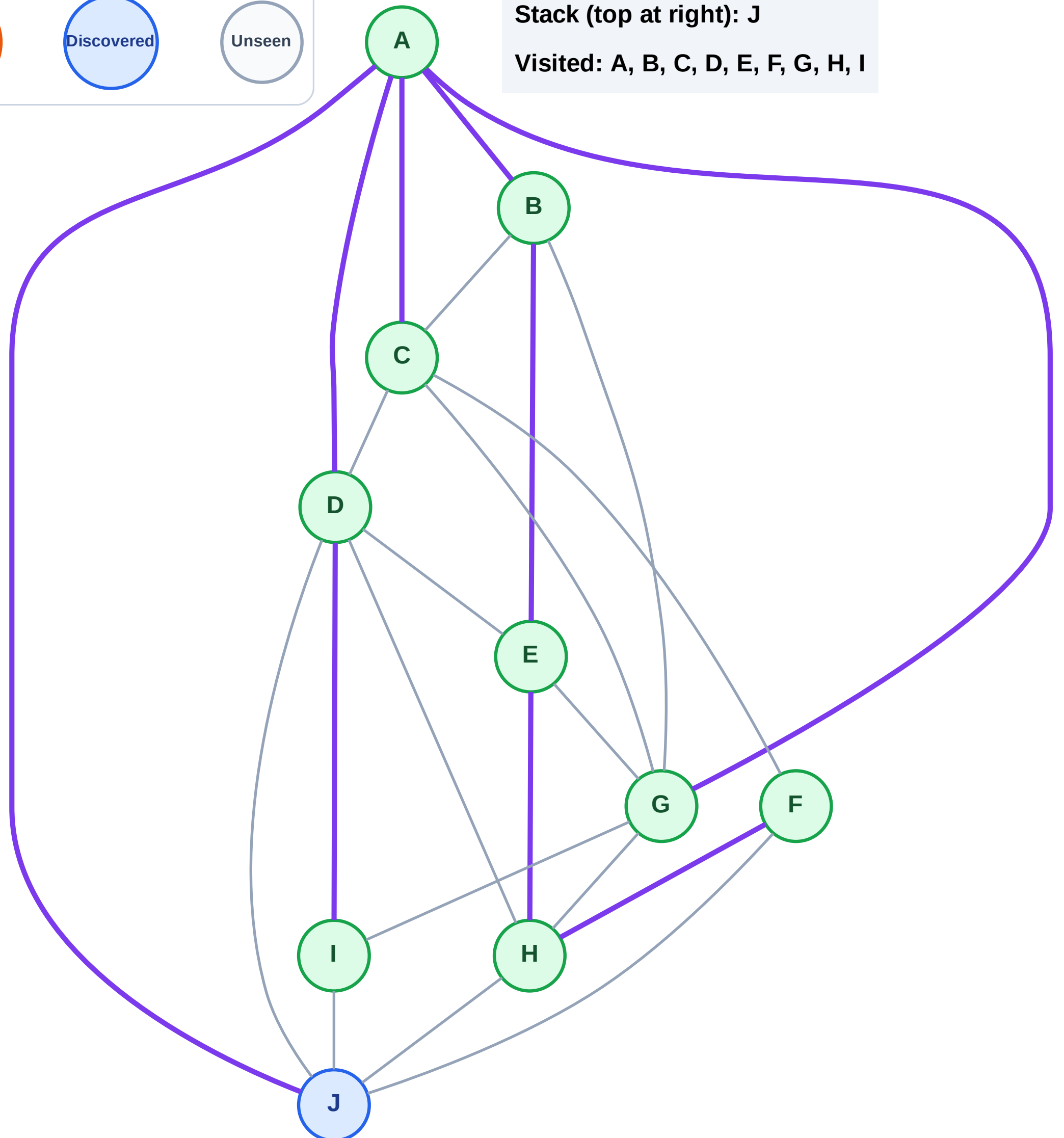
Step 19: No new neighbors from G

Legend



Stack (top at right): J

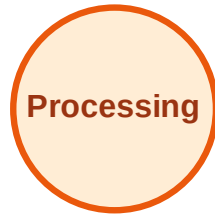
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

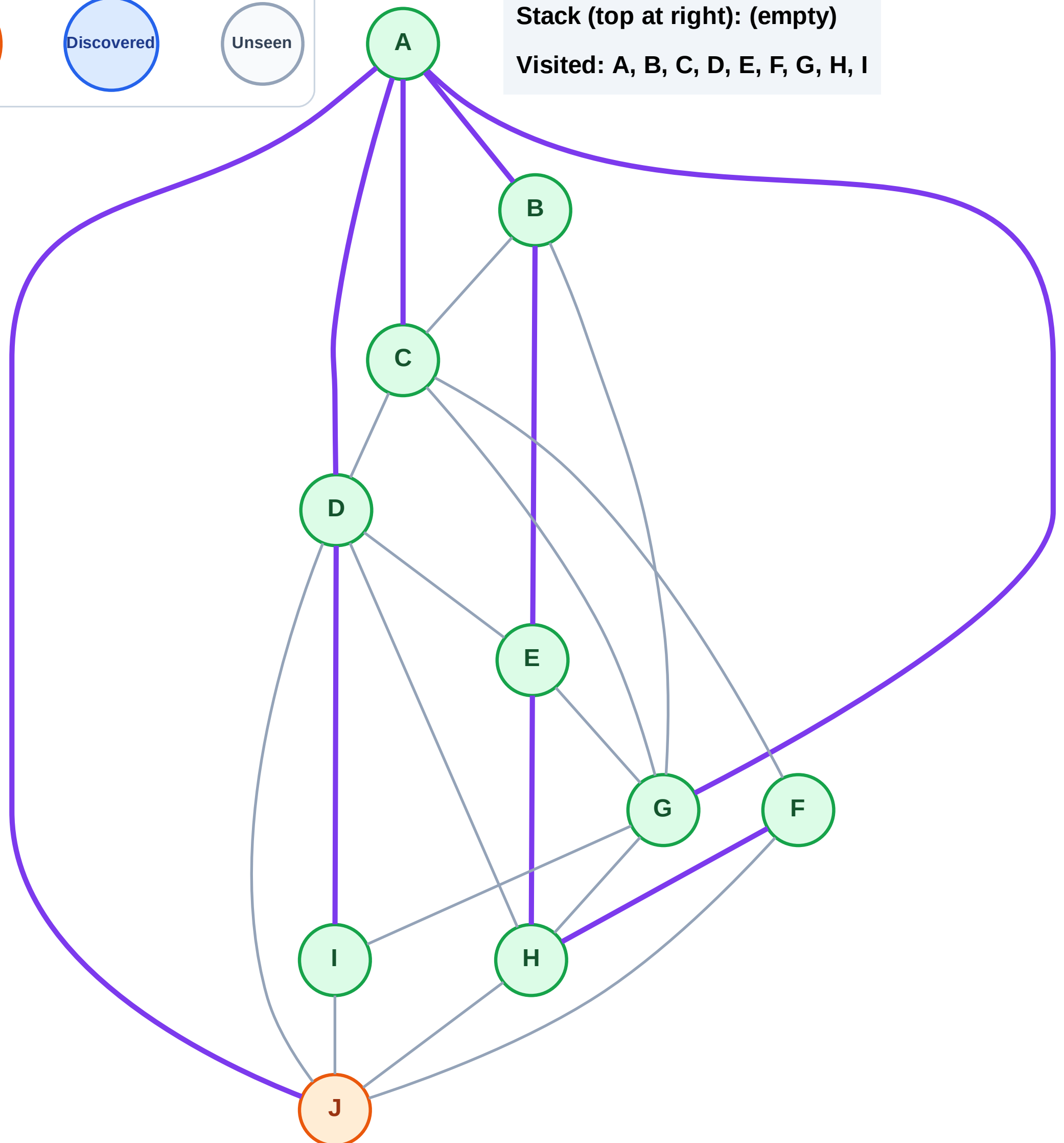
Step 20: Process node J

Legend



Stack (top at right): (empty)

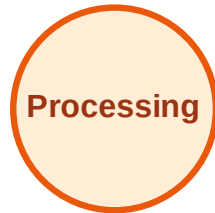
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

